Inferential Statistics Module

NAT

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Contents

1 Statistical Inference **5**

1.1 Course Learning Outcomes (CLOs) . . . 5

1.2 Assessments . . . 5

1.3 Course Structure . . . 5

2 Probability (Basic Concepts) **7**

2.1 Basic Concepts of Statistics & Random Experiments . . . 7

2.2 Definition of a Set & Membership . . . 7

2.3 Specifying sets . . . 7

2.4 Standard sets . . . 8

2.5 Definition and Properties of Probability . . . 8

2.6 Review of Set Algebra . . . 9

2.7 Additional Properties of Probability . . . 10

2.8 Method of Enumeration & Multiplication Principle . . . 12

2.9 Probability Distribution . . . 12

2.10 Permutations and Combinations (Sampling) . . . 12

2.11 Permutation Calculation Example . . . 13

2.12 Intervals . . . 13

2.13 Probability . . . 13

2.14 Independent vs. Dependent Events . . . 18

2.15 Conditional Probability . . . 26

2.16 Conditional Probability: Example . . . 29

2.17 Tutorials . . . 40

2.18 Discussion on Monty Hall Problem . . . 43

3 Chapter 2: Random Variables, Expectation, and Moments **47**

3.1 Random Variables . . . 47

3.2 RV and Distribution Function . . . 48

3.3 Discrete Random Variable . . . 49

3.4 Continuous Random Variable 51
3.5 Expected Values and Expected Function 57
3.6 Variance and standard deviation 59
3.7 Functions of Random Variable 62
3.8 Moments and Moment generating function 66
3.9 Tutorials 67

4 Random Variables and Distributions **71**

5 Overview **73**

6 Random Walk for Discrete Random Variable **75**

Chapter 1

Statistical Inference

Welcome to Inferential Statistics. This 14-week module explores probability theory, random variables, distribution functions, point estimation, confidence intervals, and hypothesis testing. We will use R to simulate and calculate statistical concepts.

1.1 Course Learning Outcomes (CLOs)

By the end of this course, students will be able to:

CLO 1: APPLY the concept of probability in relation to the concept of sampling error. (C3)

CLO 2: EXPLAIN the concept of point and confidence interval estimation for given parameters. (C2, A4)

CLO 3: DESIGN appropriate hypothesis testing based on given statistical problems. (C5, P7)

1.2 Assessments

Assessments	Percentage
Quiz 1	10%
Midterm	20%
Quiz 2	10%
Assignmnet	20%
Final	40%
Total	100%

1.3 Course Structure

Week	Topic	Assessment (Formative / Summative)
Week 1	Chapter 1: Probability Basics (Part 1)	Tutorial 1
Week 2	Chapter 1: Probability Basics (Part 2)	Tutorial 2
Week 3	Chapter 2: Expectation, Variance, and MGF	Tutorial 3 and **Quiz 1**
Week 4	Chapter 3: Expectation, Variance, and MGF	Tutorial 4

Week	Topic	Assessment (Formative / Summative)
Week 5	Chapter 3: Known Distributions (Discrete & Continuous)	Tutorial 5
Week 6	Chapter 3: Known Distributions (Discrete & Continuous)	Tutorial 6
Week 7	Chapter 4: Distribution of Function of Random Variables	Tutorial 7 and **Midterm**
Midterm	**Break**	
Week 8	Chapter 4: CLT & Approximations	Tutorial 8
Week 9	Chapter 5: Point Estimation (MLE & Method of Moments)	Tutorial 9
Week 10	Chapter 5: Point Estimation (MLE & Method of Moments)	Tutorial 10 and **Quiz 2**
Week 11	Chapter 6: Confidence Intervals	Tutorial 11
Week 12	Chapter 6: Confidence Intervals	Tutorial 12
Week 13	Chapter 7: Test of Statistical Hypotheses	Tutorial 11
Week 14	Presentation of Assignment	Past Year Paper Discussion

Chapter 2

Probability (Basic Concepts)

2.1 Basic Concepts of Statistics & Random Experiments

- Statistics involves collecting, classifying, summarizing, organizing, analyzing, and interpreting data
- A random experiment is an experiment where the outcome cannot be predicted with certainty.
- **Defining Characteristics of a Random Experiment:**

  (a) All possible outcomes of the experiment are known in advance.
  (b) Any single performance of the experiment results in an outcome that is not known in advance.
  (c) The experiment can be repeated under identical conditions.

2.2 Definition of a Set & Membership

- **Definition** A set is simply a collection of objects, these objects being called elements or members of the set.
- Examples include the set of all real numbers and the set of all UPNM students
- Membership of a set X by an object x is denoted by a special symbol $\in$, read as "belongs to"
- The statement $x \in X$ means that x is an element of, or belongs to, X .
- There is also an opposite statement $x \notin X$ which means that x is not an element of, or does not belong to, X .
- **Example:** For example, with S being the set of all students registered for TPQ3783, $\in S$ (enter your name)

2.3 Specifying sets

- Example of a set

$$S := \{1, 2, 3, 4, 5, 6\} \tag{2.1}$$

  - The notation ":= " means "is defined to be equal to
  - S is a set that contains exactly six numbers listed
  - The order of the elements does not matter. Its membership/ existance in the set is important
  - If the set is large (multiple/ many members/ elements) **and** have a clear pattern, ... is typically used. E.g.: $S := \{1, \ldots, 20\}$

2.4 Standard sets

2.5 Definition and Properties of Probability

- Probability is a real-valued set function P that assigns a number $P(A)$ to each event A in the sample space S.
- $P(A) \geq 0$
- $P(S) = 1$
- If $A_1, A_2, ...$ are mutually exclusive events ($A_i \cap A_j = \emptyset$ for $i \neq j$), then $P(A_1 \cup A_2 \cup ...) = P(A_1) + P(A_2) + ...$
- $P(\emptyset) = 0$ and $0 \leq P(A) \leq 1$
- $P(A') = 1 - P(A)$
- $P(B \cap A') = P(B) - P(B \cap A)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- If $A \subset B$, then $P(A) \leq P(B)$

2.5.1 Examples of Random Experiments

- **The experiment of tossing a coin:**
  - There are two possible outcomes: getting a Head (H) or getting a Tail (T).
  - Even though we know both **possible outcomes**, we cannot definitively predict that we will get a tail if we throw the coin in a particular manner.
- **The experiment of rolling a die:**
  - There are six possible outcomes: the die showing up 1, 2, 3, 4, 5, or 6.
  - We cannot for sure predict that we can make a particular number appear on the face of the dice when it is rolled.

2.5.2 Applied Real-Life Examples

- **Quality Control & Manufacturing:**
  - Testing the length of life of a 60-watt light bulb until it burns out.
    - There infinitely number of possible of outcomes. (Why?)
- **Environmental Testing:**
  - Counting the number of bacteria colonies in a 100-milliliter water sample taken from a specific lake basin.
- **Technology Reliability:**
  - Measuring the continuous amount of time in hours that a computer functions before breaking down.
- **Queueing & Logistics:**
  - Measuring the waiting time in minutes until the first customer or patient arrives at a facility.

2.5.3 Properties of Probability

- Sample Space, S
- The event set, E
- The probability measure, p

2.5.4 Properties of Probability: Examples

- **The experiment of tossing a coin:**
  - Sample Space, $S := \{H, T\}$
  - Event set of getting a Head, $E := \{H\}$
  - Probability measure of E: $p = \frac{1}{2}$

Probability measure	Description
$P(A) \geq 0$	Probability of an event A is 0 or greater
$P(S) = 1$	Probability of space S is 1
$0 \leq P(A) \leq 1$	Probability is always between 0 and 1.

2.6 Review of Set Algebra

Symbols	Description
$\emptyset$	The null set or empty set.
$A \subset B$	A is a subset of B.
$A \cup B$	the union of A and B, representing elements in either set.
$A \cap B$	The intersection of A and B, representing elements shared by both sets.
A'	The complement of A, representing everything outside of A.
Mutually Exclusive:	Events $A_1, A_2, \ldots, A_k$ are mutually exclusive if $A_i \cap A_j = \emptyset$ for $i \neq j$; they are disjoint sets.
Exhaustive Events:	Events $A_1, A_2, \ldots, A_k$ are exhaustive if their union forms the entire sample space, $A_1 \cup A_2 \cup \cdots \cup A_k = S$.

- So if $A_1, A_2, \ldots, A_k$ are mutually exclusive and exhaustive events, we know that $A_1 \cap A_2 = \emptyset$, for $i \neq j$ and $A_1 \cup A_2 \cup \cdots \cup A_k = S$

2.6.1 Properties of Set Operations

- **Commutative Laws:**
  - $A \cup B = B \cup A$.
  - $A \cap B = B \cap A$.
- **Associative Laws:**
  - $A \cup B \cup C = A \cup (B \cup C)$.
  - $A \cap B \cap C = A \cap (B \cap C)$.
- **Distributive Laws:**
  - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
  - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
- **De Morgan's Laws:**
  - $(A \cup B)' = A' \cap B'$. (First Law)
  - $(A \cap B)' = A' \cup B'$. (Second Law)

2.7 Additional Properties of Probability

- Probability is a real-valued set function P that assigns a number $P(A)$ to each event A in the sample space S.

Properties of probability	Description
$P(A) \geq 0$	Probability of an event A is 0 or greater.
$P(S) = 1$	Probability of space S is 1.
$P(\emptyset) = 0$	Probability of an empty set $\emptyset$ is 0.
$0 \leq P(A) \leq 1$	Probability is always between 0 and 1.
$P(A') = 1 - P(A)$	Given event A. The probability of the complement of event A, i.e. A^c or A' is $1 - P(A)$
$P(A) = \sum_{i=1}^{\infty} P(A \cap B_i)$	
$P(A) \leq P(B)$	If $A \subset B$, then $P(A) \leq P(B)$
$P(B \cap A)$	Probability that elements exists in both events A and B
* $P(B \cap A') = P(B) - P(B \cap A)$	
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	
$P(A_1 \cup A_2 \cup ...) =$ $P(A_1) + P(A_2) + ...$	If $A_1, A_2, ...$ are mutually exclusive events ($A_i \cap A_j = \emptyset$ for $i \neq j$), then $P(A_1 \cup A_2 \cup ...) = P(A_1) + P(A_2) + ...$

2.7.1 Axioms of probabilities

1.
2. If P is a probability function and A and B are any sets in $\mathcal{B}$, then

a. $P(A \cap A^c) = P(B) - P(A \cap B)$

Proof:

$$B = (B \cap A) \cup (B \cap A^c) \tag{2.2}$$
$$P(B) = P(B \cap A) + P(B \cap A^c) \tag{2.3}$$
$$P(B \cap A^c) = P(B) - P(B \cap A) \tag{2.4}$$

b. $P(A \cup A^c) = P(A) + P(B) - P(A \cap B)$

Proof:

$$A \cup B = A \cup (B \cap A^c) \tag{2.5}$$
$$P(A \cup B) = P(A) + P(B \cap A^c) \tag{2.6}$$
$$P(A \cup B) = P(A) + P(B) - P(B \cap A) \tag{2.7}$$
$$\tag{2.8}$$

3.

2.7.2 Probability Calculation Example

- **Experiment:** Tossing a coin twice.
- **Sample Space:** $S := \{HH, HT, TH, TT\}$, where each outcome is equally likely with a probability of 1/4. (Hint: *equally likely*: each elements in the set of equal chance of being selected hence each element has the probability of 1/4)
  - Let $E := \{HH, HT\}$ be the event that the first toss is a head.
  - Let $F := \{HH, TH\}$ be the event that the second toss is a head.

  Find the probability of the union $P(E \cup F)$:

  Answer (step-by-step):

  1. $P(E \cup F) = P(E) + P(F) - P(E \cap F)$.
  2. $P(E \cup F) = 1/2 + 1/2 - P(\{HH\})$.
  3. $P(E \cup F) = 1 - 1/4 = 3/4$.
  4. Alternatively, $P(E \cup F) = P(\{HH, HT, TH\}) = 3/4$.

2.7.3 Probability Calculation Example

- **Experiment:** Tossing a fair dice.
- **Sample Space:** $S := \{1, 2, 3, 4, 5, 6\}$ where each outcome is equally likely
  - Let E be the event that consist of event numbers
  - Let F be the event that consist of numbers greater than 3

  Find the probability of the union $P(E \cup F)$:

  Answer (step-by-step):

  1. Probability for each element is $\frac{1}{2}$
  2. $E := \{2, 4, 6\}$, hence $P(E) = \frac{1}{2}$
  3. $E := \{4, 5, 6\}$, hence $P(F) = \frac{1}{2}$
  4. $(E \cap F) := \{4, 6\}$, hence $P(E \cap F) = \frac{1}{3}$
  5. $P(E \cup F) = P(E) + P(F) - P(E \cap F)$.
  6. $P(E \cup F) = 1 - 1/3 = \frac{2}{3}$.
  7. Alternatively, $P(E \cup F) = P(\{H2, 4, 5, 6\}) = \frac{4}{6} = \frac{2}{3}$.

2.7.4 Probability Calculation: In-class exercise (A practice to your quiz)

** Task:** To answer the following questions, you will need to pair up with a friend.

a) You are given 10 minutes. Try to answer the following questions.

b) After 10 minutes, switch your answer with your friend. 5 minutes, review your friends work and add any missing steps/ fix the calculation if needed.

1. If $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.3$, calculate the following:

a. $P(A \cup B)$

b. $P(A \cap B')$

c. $P(A' \cup B)$

[6 mark] 2. A fair die is rolled three times. Let A_1 := Outcome is 1 or 2 on the first rolle, A_2 := Outcome is 3 or 4 on the first rolle, and A_3 := Outcome is 5 or 6 on the first rolle. Given that

- $P(A_i) = \frac{1}{3}$ for $i = 1, 2, 3$
- $P(A_i \cap A_j) = \left(\frac{1}{3}\right)^2$ for $i \neq j$ and $P(A_1 \cap A_2 \cap A_3) = \left(\frac{1}{3}\right)^3$.

Find $P(A_1 \cup A_2 \cup A_3)$.

[4 mark]

2.8 Method of Enumeration & Multiplication Principle

- The method of enumeration is used to determine the **number of outcomes associated with the events of certain random experiments**.
- **Multiplication Principle:** If an experiment E_1 has n_1 outcomes and for each possible outcome, a second experiment E_2 has n_2 possible outcomes, the composite experiment E_1E_2 has n_1n_2 possible outcomes.
- **Example:** Tossing a coin (E_1 with 2 outcomes) and rolling a die (E_2 with 6 outcomes).
  - The total number of outcomes is $2 \times 6 = 12$.
  - The sample space represented in ordered pairs: $S = \{(H,1),(H,2),(H,3),(H,4),(H,5),(H,6),(T,1),(T,2),(T,3)$
- If an experiment is repeated multiple times, such as tossing a coin ten times, the number of outcomes in the sample space is $2 \times 2 \times \cdots \times 2 = 2^{10}$.

2.9 Probability Distribution

- Experiment involves arrangement, selection and replacement.
- Suppose that we have 4 letter a,b,c and d, the number of possible four-letter code easily can be calculated if each letter can be repeated which is $4^4 = 256$
- If consider without repetition, then the number of possible four-letter code is the number of permutation of 4 letter which is $4! = 24$.

2.10 Permutations and Combinations (Sampling)

- **Arrangement and Replacement Example:** Given 4 letters a, b, c, and d.
  - With repetition: The number of possible four-letter codes is $4^4 = 256$.
  - Without repetition: The number of possible four-letter codes is a permutation of 4 letters, which is $4! = 24$.
- **Case 1 (Replacement, Ordered):** Sampling with replacement of an ordered sample of size r yields n^r arrangements, i.e. $r = n^r$.
- **Case 2 (No Replacement, Ordered - Permutation):** Sampling without replacement of r ordered items from n objects.
  - ${}_nP_r = \frac{n!}{(n-r)!}$. It must be true that $r \leq n$.

- **Case 3 (No Replacement, Unordered - Combination):** Sampling without replacement of r selected objects from n objects with unordered subsets.
  - ${}_nC_r = \frac{n!}{r!(n-r)!}$.

2.11 Permutation Calculation Example

1. How many four-letter code words are possiblle using the letters in **IOWA** if

   a. The letters may not be repeated

   - There are 4 letters.
   - Each time a letter is placed, the number available reduced by 1
   - $$4 \times 3 \times 2 \times 1 = 24$$

   b. The letters may be repeated

2. Suppose that two tennis players A and B are playing in a match. The first player that wins three sets will win the match. How many ways could this tennis match ends?

2.12 Intervals

- Consider two numbers, $a, b \in \mathbb{R}$
- The open interval is: $(a, b) := \{x \in \mathbb{R} : a < x < b\}$
- The close interval is: $[a, b] := \{x \in \mathbb{R} : a \leq x \leq b\}$
- The open and close interval: $(a, b] := \{x \in \mathbb{R} : a < x \leq b\}$
- The close and open interval: $[a, b) := \{x \in \mathbb{R} : a \leq x < b\}$
- Unbounded intervals: $[a, \infty) := \{x \in \mathbb{R} : a \leq x\}$
- Unbounded intervals: $(-\infty, b] := \{x \in \mathbb{R} : x \leq b\}$

2.13 Probability

2.13.1 Probability of an event

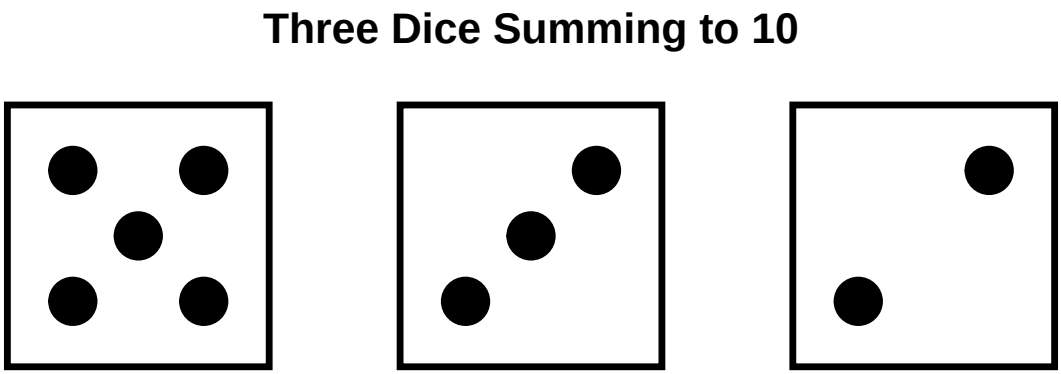
- Probability of any event, A, as the proportion of the possible outcomes which give rise to that event

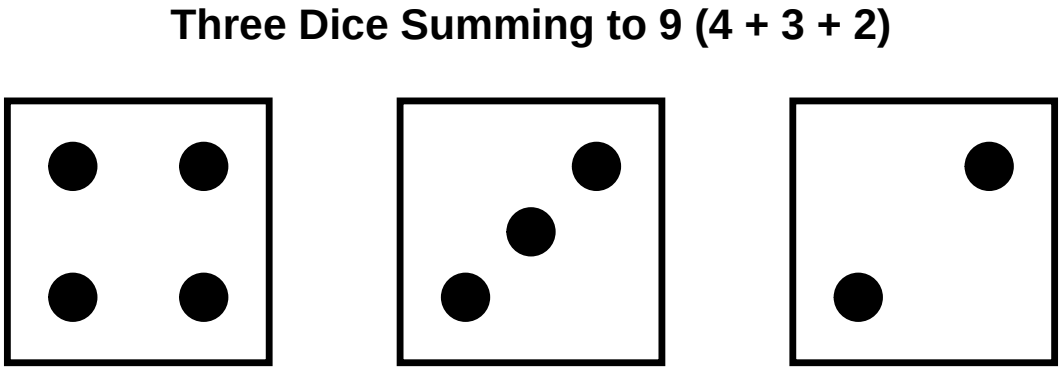
$$\frac{\text{no. of possible outcomes which give rise to event}}{\text{total no. of possible outcomes}} \tag{2.9}$$

2.13.2 Probability of an event: Equally likely outcomes

The Leibniz-Galileo Problem

In a game of tossing three fair dice, Leibniz thought that probability of getting a total is 10 and the probability that the total is 9 are equals. However, Galileo thought that these two probabilities are in the ratio 27:25.





Partition	No. of orderings	Partition	No. of orderings
Total score of 9		**Total score of 10**	
1+2+6	6	1+3+6	6
1+3+5	6	1+4+5	6
1+4+4	3	2+2+6	3
2+2+5	3	2+3+5	6
2+3+4	6	2+4+4	3
3+3+3	1	3+3+4	3
Total	**25**	**Total**	**27**

2.13.3 Probability of an event: Frequency interpretation of probability

If an "experiment" can be repeated indefinitely, then the relative frequency of any event at any stage is defined as

$$\frac{\text{no. of occurrences of event so far}}{\text{total no. of repetitions of experiment so far}} \tag{2.10}$$

- This relative frequency will obviously fluctuate randomly as the number of repetitions increases, but the expectation is that it will in the long run settle down to a limiting value which is the probability of the event.
- This method is to find the probabilities to events in cases where there is no symmetry

2.13.4 Probability of an event: Example

A batch of 15 items contains 6 defective ones. A random sample of 5 is chosen. What is the probability that 2 are defective?

Answer (Step-by-step)

1. Total no.of possible outcomes is how many ways that 5 defective items is chosen.

$\binom{15}{5} = 3003$

2. No. of possible outcomes which give rise to event

$\binom{6}{2} = 15$

3. Probability of event A: $P(A) = \frac{15}{3003}$

2.13.5 Probability of an event: Example

A batch of 15 items contains 6 defective ones. A random sample of 5 is chosen. What is the probability that **at least** 2 are defective?

Answer (Step-by-step)

1. If at least two are defective, we use OR (2 OR 3 OR 4 OR 5)

Defective	Non-defective
2	3
3	2
4	1
5	0

2. No. of possible outcomes of choosing 5 items is

$$\binom{15}{5} = 5005$$

3. The probability is

$$\frac{\binom{6}{2}\binom{9}{3} + \binom{6}{3}\binom{9}{2} + \binom{6}{4}\binom{9}{1} + \binom{6}{5}\binom{9}{0}}{3003} = \frac{2121}{3003} = 0.706$$

2.13.6 Probability of an event: Classical vs Uniform Measures

- Recall the clasical measure:
- E.g.: Let $S := \{s_1, s_2, \dots, s_k\}$. For any events $A \subset S$,

$$\frac{\text{no. of possible outcomes in } A}{\text{total no. of possible outcomes in } S} \tag{2.11}$$

- If $A := \{s_i\}$, then $P(A) =$
- For constinuous S, we use **uniform** measure
  - if S is two-dimensional and of finite area then we define

$$P(A) = \frac{\text{area of } A}{\text{area of } S}$$

- Note:
  - When an outcome is chosen at random from a sample space S: classical probability measure is used
  - When S is discrete, and the uniform measure if S is continuous.
  - For continuous probability measures, events of probability 0 are NOT impossible and events of probability 1 are NOT certain.

2.13.7 Probability of an event: Classical vs Uniform Measures: Example

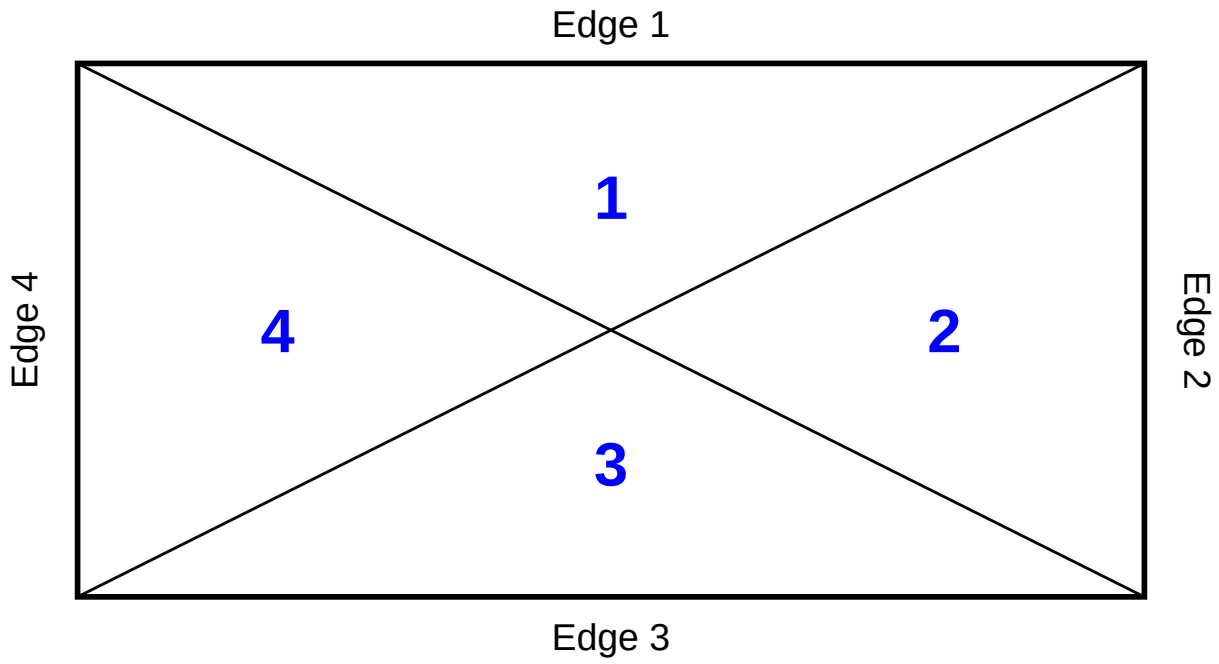
The diagram below shows a rectangle with a length of 8 and a width of 4. It is divided into four triangular regions by drawing the two diagonal lines from corner to corner. The regions are numbered from 1 to 4:

- Region 1 is the top triangle,
- Region 2 is the right triangle,
- Region 3 is the bottom triangle, and
- Region 4 is the left triangle.

The perimeter is divided into 4 edges numbered in accordance with the regions (Edge 1 is the top edge, Edge 2 is the right edge, etc.).

(i) A **point** is chosen at random from the **rectangle**. What is the probability that it is **in** region 1? What is the probability that it is **in** region 2?

(ii) A **point** is chosen at random from the **perimeter of the rectangle**. What is the probability that it is **on** edge 1? What is the probability that it is **on** edge 2?



Solution

(i) To find the probability of a point landing in a specific region, we divide the area of that region by the total area of the shape.

- Total Area: The rectangle has a length of 8 and a width of 4.

  Total Area $= 8 \times 4 = 32$.

- Region 1 (Top Triangle): The base of this triangle is the top edge of the rectangle (length 8). Because the diagonals meet exactly in the center, the perpendicular height of this triangle is half of the rectangle's width $(4 \div 2 = 2)$.

  Area of Region 1 $= \frac{1}{2} \times$ base $\times$ height $= \frac{1}{2} \times 8 \times 2 = 8$.

- Region 2 (Right Triangle): The base of this triangle is the right edge of the rectangle (width 4). The perpendicular height from the center to this edge is half of the rectangle's length $(8 \div 2 = 4)$.

  Area of Region 2 $= \frac{1}{2} \times$ base $\times$ height $= \frac{1}{2} \times 4 \times 4 = 8$.

- Note: The diagonals of any rectangle divide it into four triangles of equal area, even though they have different perimeters

- Dividing the region areas by the total area gives the final probabilities:

  - The probability that a random point is in Region 1 is

$$\frac{8}{32} = \frac{1}{4} = 0.25$$

  - The probability that a random point is in Region 2 is

$$\frac{8}{32} = \frac{1}{4} = 0.25$$

(ii) To find the probability of a point landing on a specific edge, we divide the length of that edge by the total perimeter.

- Total Perimeter: The rectangle has two sides of length 8 and two sides of length 4.Total Perimeter $= 8 + 8 + 4 + 4 = 24$.
  - Edge 1 (Top Edge): This edge has a length of 8.
  - Edge 2 (Right Edge): This edge has a length of 4.
- Dividing the edge lengths by the total perimeter gives the final probabilities:
  - The probability that a random point on the perimeter is on Edge 1 is

$$\frac{8}{24} = \frac{1}{3} = 0.333$$

  - The probability that a random point on the perimeter is on Edge 2 is

$$\frac{4}{24} = \frac{1}{6} = 0.167$$

2.13.8 Probability of an event: Classical vs Uniform Measures: Example

- Consider the experiment where one point is chosen at random from the interval $[0, 1]$. For $a \in [0, 1]$, $P(a) = 0$, that is, the event $x = a$ has probability 0, in the uniform probability measure.
  - Points have zero length.
  - But that does not mean that this event is impossible:
  - if we choose a point at random from $[0, 1]$ then any point could be chosen but the probability that a particular point is chosen is 0.
- To get a positive probability, we need an event:
  - E.g.: $\frac{1}{4} \leq a \leq \frac{3}{4}$ where the probability is $\frac{1}{2}$

2.14 Independent vs. Dependent Events

- **Problem Context:** Consider selecting 2 cards from a standard deck of 52 cards.
  - *With replacement:* The probability of obtaining two kings involves **independent events**.
  - *Without replacement:* The probability of obtaining two kings involves **dependent events**.
- **Independent Events:** Two events are independent if the occurrence of one of the events does not affect the probability of the occurrence of the other event.

$$P(A \cap B) = P(A)P(B) \tag{2.12}$$

- For more then two finite events, a set of *mutually exclusive* events are

$$P(A_1 \cap A_2 \cap ... A_k) = P(A_1)P(A_2) ... P(A_k) \tag{2.13}$$

- **Dependent Events:** Two events are dependent if the occurrence of one of the events affects the probability of the occurrence of the other event.
- Problems with dependent events can often be solved using conditional probability, also known as the multiplication rule.
- Events that are not independent are dependent.
- Event A and B are **independent** iff $P(A \cap B) = P(A \cap B) = P(A)P(B)$. Otherwise, A and B are **dependent**
- Dependent events can be solved using tree diagaram

2.14.1 Independent Events: Example

Let A and B be independent events with $P(A) = 0.7$ and $P(B) = 0.2$. Compute

a. $P(A \cap B)$

Since A and B are independent,

$$P(A \cap B) = P(A)P(B) = 0.7 \times 0.2 = 0.14 \tag{2.14}$$

b. $P(A \cup B)$

Since A and B are independent, then using the

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.7 + 0.2 - 0.14 = 0.76 \tag{2.15}$$

c. $P(A' \cup B')$

Using De Morgan's second law,

$$P(A' \cup B') = 1 - P(A \cap B)' = 1 - 0.76 = 0.24 \tag{2.16}$$

2.14.2 Independent Events: Example

Toss a fair coin twice. Let A be the event "head on first toss", B be the event "head on second toss" and C be the event "same face on both tosses". Show that these three events are pairwise independent but not mutually independent.

Solution:

1. Toss a fair coin twice lead to a sample space

$$S := \{HH, HT, TH, TT\}$$

2. The sets of event A, B and C are

$$\begin{aligned} A :=& \{HH, HT\} && (2.17)\\ B :=& \{HH, TH\} && (2.18)\\ C :=& \{HH, TT\} && (2.19)\\ & && (2.20) \end{aligned}$$

3. The probability of each events are

$$\begin{aligned} P(A) =& \frac{1}{2} && (2.21)\\ P(B) =& \frac{1}{2} && (2.22)\\ P(C) =& \frac{1}{2} && (2.23) \end{aligned}$$

4. The intersection of the events are

$$\begin{aligned} A \cap B :=& \{HH\} && (2.24)\\ A \cap C :=& \{HH\} && (2.25)\\ B \cap C :=& \{HH\} && (2.26)\\ & && (2.27) \end{aligned}$$

5. To show pariwise independent, we need to show that

$$\begin{aligned} P(A \cap B) =& P(A)P(B) && (2.28)\\ P(A \cap C) =& P(A)P(C) && (2.29)\\ P(B \cap C) =& P(B)P(C) && (2.30) \end{aligned}$$

6. Hence,

$$P(A \cap B) = \frac{1}{4} \rightarrow P(A)P(B) = \frac{1}{2}\frac{1}{2} = \frac{1}{4} \tag{2.31}$$
$$P(A \cap C) = \frac{1}{4} \rightarrow P(A)P(C) = \frac{1}{2}\frac{1}{2} = \frac{1}{4} \tag{2.32}$$
$$P(B \cap C) = \frac{1}{4} \rightarrow P(B)P(C) = \frac{1}{2}\frac{1}{2} = \frac{1}{4} \tag{2.33}$$

7. Therefore, the events A, B, and C are pairwise independent.

8. Three events A, B, and C are mutually independent if they are pairwise independent and satisfy

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

9. The intersection of events are

$$A \cap B \cap C := \{HH\}$$

so the probabilty is

$$P(A \cap B \cap C) = \frac{1}{4}$$

10. The product of probability is

$$P(A) \cdot P(B) \cdot P(C) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$$

Since $P(A \cap B \cap C) \neq P(A) \cdot P(B) \cdot P(C)$ ($\frac{1}{4} \neq \frac{1}{8}$), the events A, B, and C are not mutually independent.

2.14.3 Independent Events: Exercise

If $P(A) = 0.8$, $P(B) = 0.5$, and $P(A \cup B) = 0.9$, are A and B independent events? Show your answer.

Solution:

To show A and B are independent, we need to show $P(A)P(B) = P(A \cap B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \tag{2.34}$$
$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.8 + 0.5 - 0.9 = 0.4 \tag{2.35}$$
$$P(A)P(B) = (0.8)(0.5) = 0.4 = P(A \cap B) \tag{2.36}$$

Since $P(A \cap B) = 0.4$ and $P(A) \cdot P(B) = 0.4$, the condition $P(A \cap B) = P(A) \cdot P(B)$ is satisfied.

2.14.4 Independent Events: Exercise

Let A be the event that the sum of the two numbers shown is 7, let B be the event that the sum of the two numbers shown is 8 and C be the event that the first number shown is even.

a. Find the probabilities of the events A, B, C, $A \cap C$ and $B \cap C$.

b. Are A and C independent? Are B and C independent?

c. Justify your answers.

Solution:

(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

a. Probabilities of EventsEvent

- A (Sum is 7):
  - $A = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$
  - Number of outcomes $n(A) = 6$
  - $P(A) = \frac{6}{36} = \frac{1}{6}$
- Event B (Sum is 8):$
  - B = {(2,6), (3,5), (4,4), (5,3), (6,2)}$
  - Number of outcomes $n(B) = 5$
  - $P(B) = \frac{5}{36}$
- Event C (First number shown is even):
  - $C = \{(2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,$
  - Number of outcomes $n(C) = 18$
  - $P(C) = \frac{18}{36} = \frac{1}{2}$
- Event $A \cap C$ (Sum is 7 AND first number is even):
  - $A \cap C = \{(2,5), (4,3), (6,1)\}$
  - Number of outcomes $n(A \cap C) = 3$
  - $P(A \cap C) = \frac{3}{36} = \frac{1}{12}$
  - Event $B \cap C$ (Sum is 8 AND first number is even):
    - $B \cap C = \{(2,6), (4,4), (6,2)\}$
    - Number of outcomes $n(B \cap C) = 3$
    - $P(B \cap C) = \frac{3}{36} = \frac{1}{12}$

b. Independence Results$A$ and C are independent.B and C are NOT independent.

c. Justification

- Two events X and Y are independent if and only if $P(X \cap Y) = P(X) \cdot P(Y)$.
- For A and C:
  - $P(A \cap C) = \frac{1}{12}$
  - $P(A) \cdot P(C) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$
  - Since $P(A \cap C) = P(A) \cdot P(C)$, events A and C are independent.
- For B and C:
  - $P(B \cap C) = \frac{3}{36} = \frac{1}{12}$
  - $P(B) \cdot P(C) = \frac{5}{36} \times \frac{1}{2} = \frac{5}{72}$
  - Since $P(B \cap C) \neq P(B) \cdot P(C)$ ($\frac{6}{72} \neq \frac{5}{72}$), events B and C are not independent.

2.14.5 Independent Events vs. Dependent Events: Examples

Independent Events

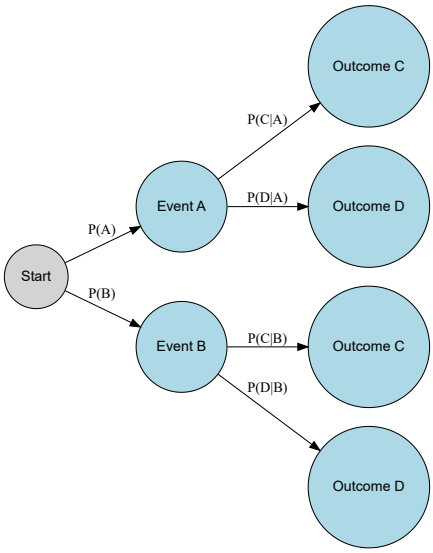
- Flipping a coin and rolling a die: Getting "Heads" on the coin doesn't change your 1-in-6 chance of rolling a "4" on the die.
- Buying a lottery ticket and getting a flat tire: Winning (or losing) the lottery has no influence on the probability of your tire popping on the way home.
- Having a boy, then having another boy: The biological odds of having a boy are roughly 50%. Having a boy first does not magically make the odds of having a girl next time 100%. The odds reset to 50% for the second child.
- When training a model to recognize tumors in MRI scans, Patient A's scan is completely independent of Patient B's scan. The outcome of one does not change the probability of the other. If you shuffle the order of the images during training, the model learns exactly the same way.
- This classification algorithm (often used for spam filtering) calculates probabilities by making a "naive" assumption: it treats every feature as an independent event. It assumes the presence of the word "urgent" in an email is completely independent of the presence of the word "money," multiplying their individual probabilities together to guess if the email is spam.

Dependent Events

- If you draw an Ace from a standard 52-card deck and keep it in your hand, the odds of drawing another Ace change. There are now only 3 Aces left in a 51-card deck.
- If you have 10 black socks and 10 white socks, the odds of pulling a black sock are 50%. If you pull out a black sock and put it on your foot, there are now only 9 black socks left out of 19. The odds for your second pull have changed.
- If there is a major accident on your route to work (Event A), the probability that you will be late to work (Event B) drastically increases. (Traffic data)
- Text generation relies entirely on dependent probability. If an LLM is generating a sentence and has already output "The quick brown", the probability of the next word being "fox" is massively higher than it would have been if the previous words were "I ate a". The next prediction depends directly on the sequence of previous predictions. (Large Language Models (NLP))
- The probability that it will rain on Tuesday is highly dependent on the barometric pressure and weather conditions from Monday. Models like ARIMA or Recurrent Neural Networks (RNNs) are specifically designed to handle this dependency. (Time Series Forecasting)

2.14.6 Probability Tree Diagram

- Tree diagrams can be used to determine the value of the probability for dependent events.


Outcome C
P(C|A)
Event A
P(D|A)
Outcome D
P(A)
Start
P(B)
Event B
P(C|B)
Outcome C
P(D|B)
Outcome D

2.15 Conditional Probability

- **Definition 1.3-1:** The conditional probability of an event A, given that event B has occurred, is defined mathematically.

- The formula for conditional probability is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

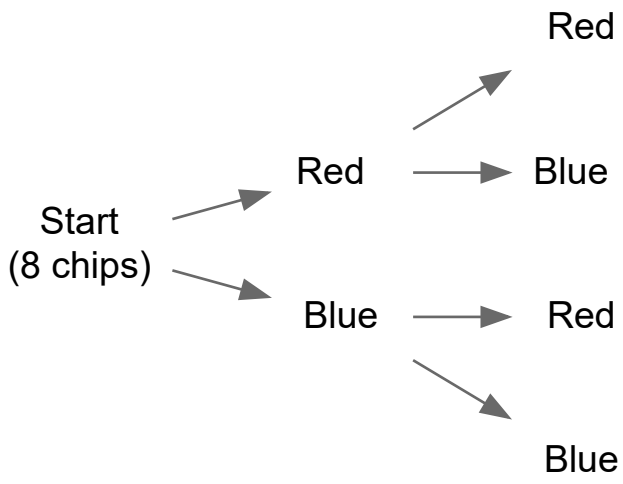
provided that $P(B) > 0$.

2.15.1 Properties of Independent Events

- **Definition:** Events A and B are independent if and only if $P(A \cap B) = P(A)P(B)$. Otherwise, A and B are called dependent events.
- **Properties of Independent Events:**
  - If A and B are independent, then $P(A|B) = P(A)$ if $P(B) > 0$, and $P(B|A) = P(B)$ if $P(A) > 0$.
  - If A and B are independent, so are A and B', A' and B, and A' and B'.

2.15.2 Conditional Probability: Example

A bowl contains 8 chips, where 3 of the chips are red and the remaining five are blue. Two chips are to be drawn successively, at random and without replacement. We use conditional probability to find the probability that the first draw results in a red chip (C_1) and that the second draw results in a blue chip (C_2).



a. Calculate the conditional probability that a red chip was drawn in the second step given that the a blue chip was drawn previously.

b. What is the probability of drawing a red chips in the second time?

Solution:

a. When a blue chip is drawn on the first try (without replacement):

- The total remaining chips in the bowl becomes $8 - 1 = 7$.
- The remaining number of red chips is still 3.
- Therefore, the conditional probability is:

$$P(R_2 \mid B_1) = \frac{3}{7}$$

b. Probability of drawing a red chip on the 2nd draw: $P(R_2)$

- Using the Law of Total Probability, drawing a red chip on the second draw can happen under two mutually exclusive scenarios depending on the outcome of the first draw:
- First draw is Red, second draw is Red ($R_1 \cap R_2$)
- First draw is Blue, second draw is Red ($B_1 \cap R_2$)

$$P(R_2) = P(R_1 \cap R_2) + P(B_1 \cap R_2)$$

$$P(R_2) = P(R_1) \cdot P(R_2 \mid R_1) + P(B_1) \cdot P(R_2 \mid B_1)$$

- Substitute the probabilities:
  - $P(R_1) = \frac{3}{8}$
  - $P(R_2 \mid R_1) = \frac{2}{7}$ (if a red is drawn first, 2 red chips remain out of 7)
  - $P(B_1) = \frac{5}{8}$
  - $P(R_2 \mid B_1) = \frac{3}{7}$

$$P(R_2) = \left(\frac{3}{8} \times \frac{2}{7}\right) + \left(\frac{5}{8} \times \frac{3}{7}\right)$$

$$P(R_2) = \frac{6}{56} + \frac{15}{56} = \frac{21}{56} = \frac{3}{8}$$

Conclusion: The probability of drawing a red chip on the second draw is $\frac{3}{8}$ (or 0.375).

2.15.3 The Birthday Problem

What is the probability that, in a group of n people, two of them share the same birthday?

How many people would need to be in a room for there to be a 50 percent chance that at least two of them share the same birthday?

- **Person 1:** Enters an empty room with 365 out of 365 days available.
- **Person 2:** Given Person 1 claimed a day, they only have 364 safe days left to avoid a match ($P = \frac{364}{365}$).
- **Person 3:** Strictly dependent on the first two outcomes, they only have 363 days left ($P = \frac{363}{365}$).

- **Person 25:** Strictly dependent on the first 24 outcomes, they only have 341 safe days left ($P = \frac{341}{365}$).
- To find the probability of at least one shared birthday, we multiply the conditional probabilities of *avoiding* a match, then use the rule of complements.

$$P(\text{Match}) = 1 - \left(\frac{365}{365} \times \frac{364}{365} \times \cdots \times \frac{341}{365}\right)$$

$$P(\text{Match}) = 1 - 0.4313 = 0.5687$$

- With **23 people**, the probability crosses 50.7%.
- With **70 people**, the probability hits 99.9%.

2.15.4 Interactive Game: The Impossible Bet

- **The Setup:** Offer a crowd of 30+ people a $10 bet that a shared birthday exists in the room.
- **The Execution:** Call out months one by one. Have people born in that month stand up and state their day.
- **The Reveal:** The audience falls for the intuitive trap, but the math is heavily in your favor. Statistically, two people standing will call out the same number well before you reach December.

2.16 Conditional Probability: Example

1. When Netflix or Amazon recommends something to you, they are rarely looking at items in a vacuum. They are calculating conditional probabilities based on user behavior.

   E.g.: "What is the probability this user will watch Stranger Things (*Event A*), given that they already watched The Umbrella Academy (*Event B*)?"

2. A Markov Model is a system that transitions from one state to another, where the probability of the next state depends only on the current state.

   E.g.: "Given that the current word is 'New' (*Event B*), what is the probability that the next word is 'York' (*Event A*) versus 'shoes'?"

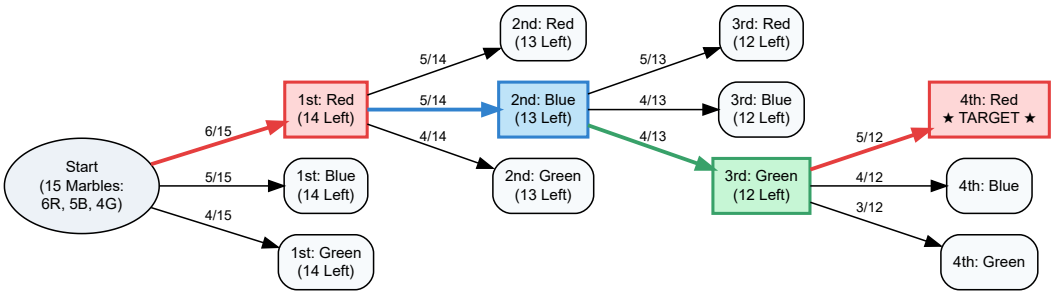
2.16.1 Conditional Probability: Exercise

A velvet bag contains 15 marbles in three different colors (three possible events per draw):

- 6 Red marbles
- 5 Blue marbles
- 4 Green marbles

You draw 4 marbles in a row without putting any back (four dependent steps).

What is the exact probability of drawing a Red, then a Blue, then a Green, and finally another Red marble?



To find the probability of drawing Red (R_1), then Blue (B_2), then Green (G_3), and finally Red (R_4) without

replacement, we use the General Multiplication Rule for dependent events:

$$P(R_1 \cap B_2 \cap G_3 \cap R_4) = P(R_1) \cdot P(B_2 \mid R_1) \cdot P(G_3 \mid R_1 \cap B_2) \cdot P(R_4 \mid R_1 \cap B_2 \cap G_3)$$

- Step 1: First Draw — Red (R_1)
  - Bag contents: 6 Red, 5 Blue, 4 Green (15 total)

$$P(R_1) = \frac{6}{15} = \frac{2}{5}$$

- Step 2: Second Draw — Blue (B_2)Bag contents after R_1: 5 Red, 5 Blue, 4 Green (14 total)

$$P(B_2 \mid R_1) = \frac{5}{14}$$

- Step 3: Third Draw — Green (G_3)Bag contents after R_1, B_2: 5 Red, 4 Blue, 4 Green (13 total)

$$P(G_3 \mid R_1 \cap B_2) = \frac{4}{13}$$

- Step 4: Fourth Draw — Red (R_4)Bag contents after R_1, B_2, G_3: 5 Red, 4 Blue, 3 Green (12 total)

$$P(R_4 \mid R_1 \cap B_2 \cap G_3) = \frac{5}{12}$$

- Final Calculation
  - Multiply all individual conditional probabilities together:

$$P(R_1 \cap B_2 \cap G_3 \cap R_4) = \frac{6}{15} \times \frac{5}{14} \times \frac{4}{13} \times \frac{5}{12}$$

$$P(R_1 \cap B_2 \cap G_3 \cap R_4) = \frac{6 \times 5 \times 4 \times 5}{15 \times 14 \times 13 \times 12} = \frac{600}{32,760}$$

  - Simplifying the fraction by dividing numerator and denominator by 120:

$$\mathbf{P(R_1 \cap B_2 \cap G_3 \cap R_4)} = \mathbf{\frac{5}{273}} \approx 0.018315 \quad \text{(or } 1.8315\%\text{)}$$

2.16.2 Conditional Probability: Exercise

An IT specialist is analysing traffic on a highly secure corporate email server. Because the security is very tight, spam is quite rare.

Only 5% of all incoming emails are actually Spam. The other 95% are Legitimate internal emails. You are looking at how often the word "URGENT" is used in these emails. Spammers love this word. If an email is Spam, there is a 40% chance it contains the word "URGENT". Normal employees rarely use it. If an email is Legitimate, there is only a 2% chance it contains the word "URGENT".

An alert pops up: A new email has just arrived, and it contains the word "URGENT". Based on conditional probability, what is the approximate percentage chance that this specific email is actually Spam?

Solution:

Let

- S: Incoming emails that are actually spam
- S': Incoming emaila that are legitimate
- U: Emails with URGENT
- U': Emails with no URGENT

Then,

- $P(S) = 0.05$
- $P(S') = 0.95$
- $P(U|S) = 0.40$
- $P(U|S') = 0.02$

To find the probability that the email is Spam given that there is URGENT in it

$$P(S|U) = \frac{P(U \cap S)}{P(U)} \tag{2.37}$$

$$= \frac{P(U|S)P(S)}{P(U)} \tag{2.38}$$

$$= \frac{0.4 \times 0.05}{P(U)} \tag{2.39}$$

where

$$P(U) = P(U|S)P(S) + P(U|S')P(S') \tag{2.40}$$

$$= 0.4 \times 0.05 + 0.02 \times 0.95 = 0.039 \tag{2.41}$$

Hence,

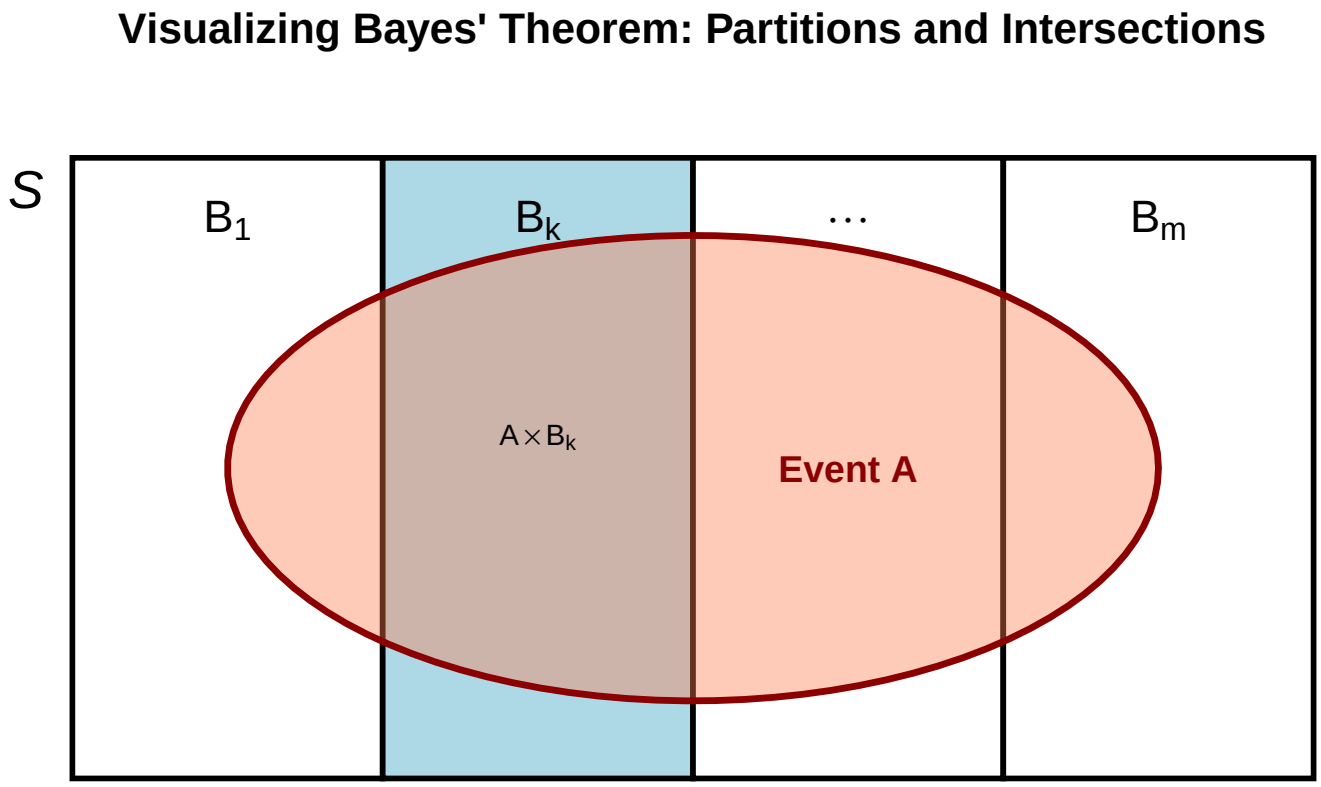
$$P(S|U) = \frac{0.4 \times 0.05}{0.039} = 0.0513 \tag{2.42}$$

Therefore, the approximate percentage that this specific email is Spam is 5.13%.

2.16.3 Bayes' Theorem

- Let $B_1, B_2, \ldots, B_m$ constitute a partition of the sample space S.
- This means that $S = B_1 \cup B_2 \cup \cdots \cup B_m$ and $B_i \cap B_j = \emptyset$ for $i \neq j$.
- Then for any event A that is a subset of S, Bayes' Theorem is given by:

$$P(B_k|A) = \frac{P(B_k)P(A|B_k)}{\sum_{i=1}^{m} P(B_i)P(A|B_i)}$$



Out of the entire area where Event A happens (the complete coral oval), what exact percentage of that area belongs to our target scenario B_k?

- The numerator ($P(B_k)P(A|B_k)$) is just the area of the specific intersection chunk inside B_k.
- The denominator ($\sum_{i=1}^{m} P(B_i)P(A|B_i)$) is just the total area of the coral oval, calculated by adding up all of its individual chunks across every partition.

2.16.4 Bayes' Theorem: Proof

By the definition of conditional probability, the probability of event B_k given event A is:

$$P(B_k|A) = \frac{P(B_k \cap A)}{P(A)}$$

Using the multiplication rule of probability, we can express the intersection in the numerator as:

$$P(B_k \cap A) = P(B_k)P(A|B_k)$$

Next, because $B_1, B_2, \ldots, B_m$ form a partition of the sample space S, we can apply the Law of Total Probability to expand the denominator $P(A)$. The event A can occur in conjunction with any of the mutually exclusive events B_i:

$$P(A) = \sum_{i=1}^{m} P(B_i \cap A)$$

Applying the multiplication rule to each term in the sum gives:

$$P(A) = \sum_{i=1}^{m} P(B_i)P(A|B_i)$$

Finally, substituting the expanded numerator and the expanded denominator back into the original conditional probability formula yields Bayes' Theorem:

$$P(B_k|A) = \frac{P(B_k)P(A|B_k)}{\sum_{i=1}^{m} P(B_i)P(A|B_i)}$$

2.16.5 Bayes Rule: Example 1

- Sample Space (S): The entire population being tested.
- The Partitions (B_1 and B_2):
  - B_1 (Has Disease): 1% of the population actually has the disease.
  - B_2 (Healthy): 99% of the population does not have the disease.
- Event A (Positive Test Result): The test comes back Positive.
  - If a patient has the disease (B_1), the test is 90% accurate.
  - If a patient is healthy (B_2), the test has a 5% false positive rate

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2.16.6 Bayes Rule: Example 2 (Heart Disease (Medical Screening))

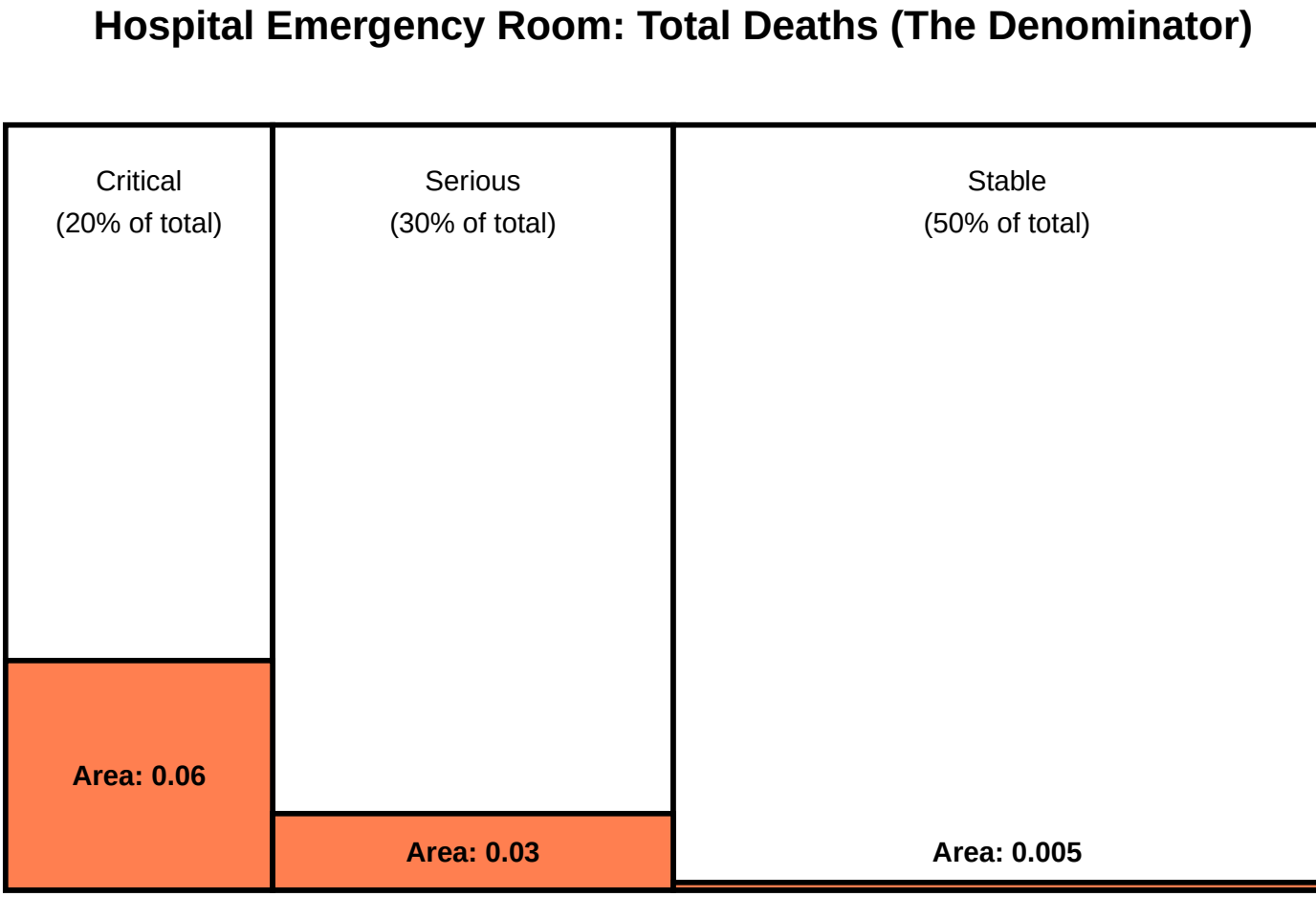
- Sample Space (S): All patients walking into a clinic.
- Partitions (B_1 and B_2):
  - B_1: Patients who actually have Heart Disease.
  - B_2: Patients who are Healthy (No Heart Disease).
- Event A (Chest Pain): A patient presents with chest pain.
  - $A \cap B_1$ (True Positive Indicator): Patients who have heart disease and experience chest pain.
  - $A \cap B_2$ (False Alarm): Healthy patients who experience chest pain due to harmless causes (like acid reflux or muscle strain).
- How it works:When a doctor sees a patient with chest pain (A), they need to determine if it is caused by heart disease (B_1). Even though chest pain is common in heart disease, acid reflux is so widespread in the general healthy population (B_2) that chest pain alone doesn't instantly mean a heart attack—it depends on the relative sizes of both groups.

2.16.7 Bayes' Theorem Application: Example

At a hospital's emergency room, patients are classified:

- 20% of them are critical,
- 30% are serious, and
- 50% are stable.

Of the critical ones, 30% die; of the serious, 10% die; and of the stable, 1% die. Given that a patient dies, what is the conditional probability that the patient was classified as critical?



a. This diagram illustrates the Law of Total Probability applied to hospital emergency room patient outcomes

b. The partitions are the vertical columns that divide the entire patient population into three mutually exclusive groups based on their initial condition upon arrival:

- B_1 = Critical Patients: Make up 20% (0.20) of all incoming patients.
- B_2 = Serious Patients: Make up 30% (0.30) of all incoming patients.

- B_3 = Stable Patients: Make up 50% (0.50) of all incoming patients.
- (Notice that 20% + 30% + 50% = 100%, meaning these three categories cover every single patient arriving at the ER).

c. 0.06 (or 6%) represents the joint probability that a patient both arrives in Critical condition AND dies.

- Mathematically, it is the area of that specific shaded box:
-
$$P(\text{Critical} \cap \text{Death}) = P(\text{Critical}) \times P(\text{Death} \mid \text{Critical}) = 0.06$$
- Since 20% (0.20) of patients are Critical, a shaded area of 0.06 means that 30% of Critical patients end up dying ($0.20 \times 0.30 = 0.06$).

Solution:

To find the probability that a patient who died was classified as critical, we apply **Bayes' Theorem**.

1. Define the Events and Given Probabilities

- **Classification probabilities:**
- $P(C) = 0.20$ (Critical)
- $P(S) = 0.30$ (Serious)
- $P(St) = 0.50$ (Stable)
- **Conditional probabilities of death (D):**
- $P(D \mid C) = 0.30$
- $P(D \mid S) = 0.10$
- $P(D \mid St) = 0.01$

2. Find the Total Probability of Death $P(D)$

Using the Law of Total Probability:

$$P(D) = P(C) \cdot P(D \mid C) + P(S) \cdot P(D \mid S) + P(St) \cdot P(D \mid St)$$

$$P(D) = (0.20 \times 0.30) + (0.30 \times 0.10) + (0.50 \times 0.01)$$

$$P(D) = 0.060 + 0.030 + 0.005 = 0.095$$

3. Calculate $P(C \mid D)$ using Bayes' Theorem

$$P(C \mid D) = \frac{P(C) \cdot P(D \mid C)}{P(D)}$$

$$P(C \mid D) = \frac{0.20 \times 0.30}{0.095} = \frac{0.060}{0.095} = \frac{12}{19}$$

$$\mathbf{P(C \mid D) = \frac{12}{19} \approx 0.6316} \quad (\text{or } \mathbf{63.16}\%)$$

2.16.8 Bayes' Theorem Application: Example

You are building a model to detect spam emails. Your historical data shows that exactly 5% of all login attempts on your platform are fraudulent. Consequently, 95% are legitimate.Your model evaluates logins based on three distinct features:

i. F_1 (Late Night): The login occurs between 2 AM and 5 AM.

ii. F_2 (VPN Used): The login IP is routed through a known commercial VPN.

iii. F_3 (Crypto Purchase): The first action after login is attempting to buy cryptocurrency.

Based on your training data, you have the following conditional probabilities likelihoods) for how often these features appear in both classes:

If a login is Fraudulent:

- $P(\text{Late Night}|\text{Fraud}) = 60\%$
- $P(\text{VPN}|\text{Fraud}) = 80\%$
- $P(\text{Crypto}|\text{Fraud}) = 50\%$

If a login is Legitimate:

- $P(\text{Late Night}|\text{Legitimate}) = 10\%$
- $P(\text{VPN}|\text{Legitimate}) = 10\%$
- $P(\text{Crypto}|\text{Legitimate}) = 20\%$

What is the posterior probability that a login is Fraudulent given that all three features (F_1: Late Night, F_2: VPN, F_3: Crypto Purchase) are present?

Solution:

Bayes' Rule Formal Definition

Bayes' Rule states that for any two events A and B, the conditional probability of A given B is:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

Application to Fraud Detection

Replacing A with Fraud and B with the observed evidence $E = (F_1 \cap F_2 \cap F_3)$:

$$P(\text{Fraud} \mid E) = \frac{P(E \mid \text{Fraud}) \cdot P(\text{Fraud})}{P(E)}$$

1. Likelihood Expansion (Naïve Bayes Assumption)

Assuming features F_1, F_2, and F_3 are conditionally independent given the class:

$$P(E \mid \text{Fraud}) = P(F_1 \mid \text{Fraud}) \cdot P(F_2 \mid \text{Fraud}) \cdot P(F_3 \mid \text{Fraud}) = 0.60 \cdot 0.80 \cdot 0.50 = 0.24$$

$$P(E \mid \text{Legitimate}) = P(F_1 \mid \text{Legitimate}) \cdot P(F_2 \mid \text{Legitimate}) \cdot P(F_3 \mid \text{Legitimate}) = 0.10 \cdot 0.10 \cdot 0.20 = 0.002$$

2. Evidence Expansion (Law of Total Probability)

$$P(E) = P(E \mid \text{Fraud}) \cdot P(\text{Fraud}) + P(E \mid \text{Legitimate}) \cdot P(\text{Legitimate})$$

$$P(E) = (0.24 \cdot 0.05) + (0.002 \cdot 0.95) = 0.012 + 0.0019 = 0.0139$$

---

3. Full Expanded Formula with Numbers

Combining all components into a single Bayes' Rule expression with values substituted:

$$P(\text{Fraud} \mid F_1, F_2, F_3) = \frac{0.60 \cdot 0.80 \cdot 0.50 \cdot 0.05}{(0.60 \cdot 0.80 \cdot 0.50 \cdot 0.05) + (0.10 \cdot 0.10 \cdot 0.20 \cdot 0.95)}$$

$$P(\text{Fraud} \mid F_1, F_2, F_3) = \frac{0.012}{0.012 + 0.0019} = \frac{0.012}{0.0139} \approx 0.8633 \quad (86.33\%)$$

2.16.9 Bayes' Theorem Application: Exercise

An influencer marketing agency is analyzing a client's Instagram account. Based on historical data, the client chooses to post either a **Story** or a **Reel**.

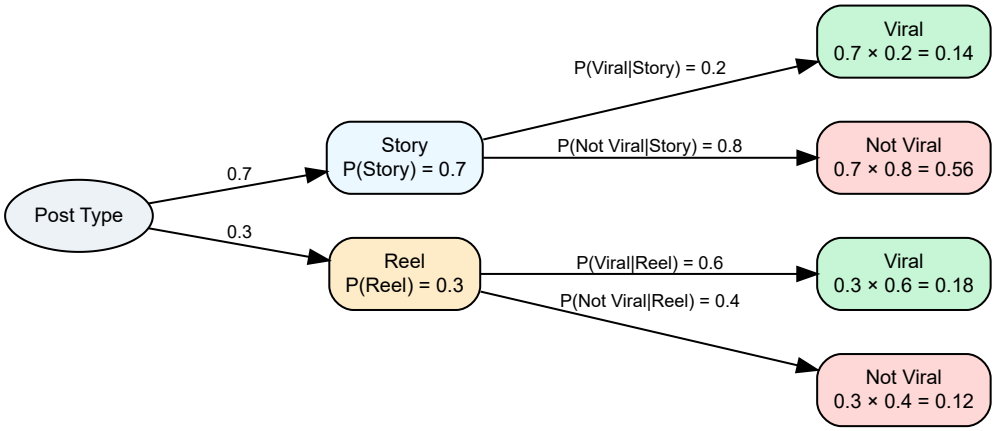
- The probability that the client posts a Story is 70% (0.7), and the probability they post a Reel is 30% (0.3).
- If they post a Story, the probability it goes "Viral" (gets over 10,000 views) is 20% (0.2).
- If they post a Reel, the probability it goes "Viral" is 60% (0.6).

a. **Draw a tree diagram** to represent this scenario. Be sure to label all branches with their respective events and probabilities.

b. Calculate the probability that a randomly selected post goes Viral.

c. **Evaluate:** If you observe that a post has gone Viral, what is the conditional probability that it was a Reel?

Solution:



a. Tree Diagram

b. Probability that a randomly selected post goes ViralUsing the Law of Total Probability:

$$P(\text{Viral}) = P(\text{Story} \cap \text{Viral}) + P(\text{Reel} \cap \text{Viral})$$

$$P(\text{Viral}) = P(\text{Story}) \cdot P(\text{Viral} \mid \text{Story}) + P(\text{Reel}) \cdot P(\text{Viral} \mid \text{Reel})$$

$$P(\text{Viral}) = (0.7 \times 0.2) + (0.3 \times 0.6)$$

$$P(\text{Viral}) = 0.14 + 0.18 = 0.32$$

Answer: The probability that a post goes Viral is 0.32 (or 32%).

c. Conditional Probability that a Viral post was a ReelUsing Bayes' Theorem:

$$P(\text{Reel} \mid \text{Viral}) = \frac{P(\text{Reel} \cap \text{Viral})}{P(\text{Viral})}$$

$$P(\text{Reel} \mid \text{Viral}) = \frac{P(\text{Reel}) \cdot P(\text{Viral} \mid \text{Reel})}{P(\text{Viral})}$$

$$P(\text{Reel} \mid \text{Viral}) = \frac{0.3 \times 0.6}{0.32} = \frac{0.18}{0.32} = \frac{18}{32} = \frac{9}{16} = 0.5625$$

Given that a post has gone Viral, the conditional probability that it was a Reel is $\frac{9}{16} = 0.5625$ (or 56.25%).

2.17 Tutorials

1. Given $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$. CALCULATE the following:

   a. $P(A \cup B)$ [2 mark]
   b. $P(A \cap B')$ [2 mark]
   c. $P(A' \cup B')$ [2 mark]

2. Let $S := \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A := \{1, 3, 5, 7, 9\}$, $B := \{1, 2, 4, 6, 7, 9\}$ and $C := \{s \in S : 2s \in B\}$. DETERMINE

   a. $A \cup B$ [2 mark]
   b. $(A \cup B)^c$ [2 mark]
   c. $C \notin A$ [2 mark]

3. Roll a fair six-sided die three times. Let

   - $A_1 = \{1 \text{ or } 2 \text{ on the first roll}\}$
   - $A_2 = \{3 \text{ or } 4 \text{ on the second roll}\}$
   - $A_3 = \{5 \text{ or } 6 \text{ on the third roll}\}$.

   It is given that

   - $P(A_i) = 1/3$ for $i = 1, 2, 3$;
   - $P(A_i \cap A_j) = (1/3)^2$ for $i \neq j$;
   - $P(A_1 \cap A_2 \cap A_3) = (1/3)^3$.

a. Find $P(A_1 \cup A_2 \cup A_3)$.[2 mark]

b. Show that $P(A_1 \cup A_2 \cup A_3) = 1 - \left(1 - \frac{1}{3}\right)^3$. [3 mark]
(Hint: You need to **show** how to get the RHS from teh LHS)

4. How many four-letter code words are possible using the letters in IOWA if:

a. The letters may not be repeated?
b. The letters may be repeated?

5. Suppose that Novak Djokovic and Roger Federer are playing a tennis match in which the first player to win three sets wins the match. Using D and F for the winning player of a set, in how many ways could this tennis match end?

6. A student uses Gemini to write a block of R code for their statistics assignment. The probability that the generated code runs perfectly without any errors on the first try is 0.82. What is the probability that the code produces at least one error on the first try? [2 mark]

7. A user prompts Gemini to generate a meal plan. The response will feature exactly one primary protein source. The probability it features chicken (C) is 0.45, and the probability it features tofu (T) is 0.20. Since a single response cannot feature both as the primary source, these events are mutually exclusive. Calculate $P(C \cup T)$, the probability that the meal plan features either chicken or tofu. [2 mark]

8. A researcher asks Gemini to analyze a dataset. Let A be the event that Gemini successfully reads the uploaded data, and let B be the event that Gemini generates a correct visualization. The probability that Gemini reads the data is $P(A) = 0.90$. Given that it reads the data, the conditional probability that it generates a correct visualization is $P(B|A) = 0.70$. Find the probability that Gemini successfully reads the data **and** generates a correct visualization. [3 marks]

9. Two fair dice are rolled. **Calculate** the conditional probability that at least one die lands on 5 given that the other die lands on different numbers? [3 mark]

10. Let $n \in \mathbb{N}$, and $r = 0, 1, 2, \ldots, n$. Show that

$$\binom{n}{r} = \binom{n}{n-r}$$

11. Two fair dice are rolled. **Calculate** the conditional probability that at least one die lands on 5 given that the other die lands on different numbers? [3 mark]

12. From historical data, 97% babies survive during delivery. 15% are from cesarean section. When a C-section is performed, the baby survives 96% of the time. If a randomly chosen pregnant woman does not have a section, whar is the probability the baby survive? [3 mark]

13. Forty-eight percent (48%) of the students of a certain university are males. Five percent (5%) of the students of this university are majoring in Mathematics. Five percent (5%) of the male students are majoring in Mathematics. If a student is selected at random, calculate the conditional probability that

a. this student is male, given that the student is majoring in Mathematics
b. this student is majoring in Mathematics, given the student is a male

14. An influencer marketing agency is analyzing a client's Instagram account. Based on historical data, the client chooses to post either a **Story** or a **Reel**. (Level 4)

- The probability that the client posts a Story is 70% (0.7), and the probability they post a Reel is 30% (0.3).

- If they post a Story, the probability it goes "Viral" (gets over 10,000 views) is 20% (0.2).
- If they post a Reel, the probability it goes "Viral" is 60% (0.6).

a. **Draw a tree diagram** to represent this scenario. Be sure to label all branches with their respective events and probabilities.

b. Calculate the probability that a randomly selected post goes Viral.

c. **Evaluate:** If you observe that a post has gone Viral, what is the conditional probability that it was a Reel?

15. A university tracks how students use Gemini for their Probability and Statistics assignments. They find that 60% of students use Gemini primarily for coding help in R (C), while 40% use it primarily for conceptual explanations (E). (Level 5)

    At the end of the semester, 80% of the students who used it for coding achieved an 'A' grade (A), whereas 90% of those who used it for conceptual explanations achieved an 'A' grade.

    a. Draw a tree diagram to represent this scenario. [2 marks]

    b. Calculate the total probability that a randomly selected student achieves an 'A' grade. [2 marks]

    c. If a randomly selected student achieved an 'A' grade, use Bayes' Theorem to find the probability that they used Gemini primarily for coding help. [3 marks]

16. You are looking at the relationship between heavy vaping and a specific, rare throat mutation that is a known precursor to cancer.Because the clinic sees a general population of young adults, the mutation is still relatively rare overall:

    - Only 2% of all patients who walk through the doors actually have this mutation.
    - The other 98% of patients have completely healthy throat tissue.

    You then look at the vaping habits of these patients:Vaping is heavily linked to the mutation.

    - If a patient has the mutation, there is a 90% chance they are a heavy vaper.
    - However, vaping is also just a common habit. If a patient is perfectly healthy, there is still a 20% chance they are a heavy vaper.

    A nurse hands you the intake file for a brand new patient in the waiting room. You haven't seen their test results, but you see one checked box: The patient is a heavy vaper. Based on conditional probability, **calculate** is the exact percentage chance that this specific patient has the throat mutation?

17. You are a contestant on a game show. The host stands in front of three closed doors: Door 1, Door 2, and Door 3. Behind one of the doors is a brand new sports car. Behind the other two doors are goats. The host asks you to pick a door. You choose Door 1.

    The host knows exactly what is behind every door. After you make your pick, the host must open one of the two remaining doors that you did not pick, and he must always reveal a goat.

    The host walks over and opens Door 3. A goat is sitting behind it. He then turns to you and asks: "Do you want to stick with your original choice (Door 1), or do you want to switch to the other closed door (Door 2)?"

    Based on conditional probability, what are your exact percentage chances of winning the car if you stick with Door 1, versus if you switch to Door 2?

2.18 Discussion on Monty Hall Problem

The Origin Story: *Let's Make a Deal*

Long before it became a staple of statistics classes, the Monty Hall problem was a genuine pop-culture phenomenon born from American daytime television.

The problem is named after **Monty Hall**, the charismatic original host of the game show ***Let's Make a Deal***, which premiered in 1963. The show was famous for its raucous studio audience—contestants dressed up in outrageous costumes (from giant bananas to medieval knights) just to get Monty's attention and a chance to play.

The climax of the show often involved the "Big Deal of the Day." Monty would present a contestant with three numbered doors. Behind one was a spectacular prize (like a brand-new car or a dream vacation). Behind the other two were "Zonks"—gag prizes like a giant pile of mismatched shoes, a live farm animal, or, famously, a goat.

The Man Who Knew Everything: The mathematical twist of the game show relies on one crucial fact: *Monty Hall always knew what was behind the doors.* He never opened the winning door by accident. He deliberately built suspense by revealing a Zonk, forcing the contestant into a psychological showdown.

2.18.1 The 1990 Math Meltdown

The game show scenario transitioned into mathematical infamy in September 1990. A reader submitted the door problem to **Marilyn vos Savant**—who held the Guinness World Record for the highest IQ—for her "Ask Marilyn" column in *Parade* magazine.

Marilyn correctly answered that the contestant should always switch doors.

The public response was explosive. She received over 10,000 letters, including nearly 1,000 from PhDs and math professors, vehemently arguing that she was wrong and that the odds were obviously 50/50.

"You blew it, and you blew it big!" wrote one mathematician. *"May I suggest that you obtain and refer to a standard textbook on probability before you try to answer a question of this type again?"* wrote another.

Marilyn stood her ground, ran mathematical simulations, and eventually proved the PhDs wrong. Today, the Monty Hall problem remains one of the most famous examples of how human intuition spectacularly fails at conditional probability.

2.18.2 The Monty Hall Problem: To Switch or Not to Switch?

Imagine you are on a game show, staring down three closed doors.

Behind one door is a brand-new car. Behind the other two? Goats.

- **Step 1:** You pick a door (let's say Door 1).
- **Step 2:** The host, Monty Hall, who knows exactly what is behind every door, opens one of the other doors (let's say Door 3) to reveal a bleating goat.
- **Step 3:** Monty turns to you and asks, *"Do you want to stick with Door 1, or switch to Door 2?"*

What do you do?

> **The Human Intuition:** "There are two doors left. One has a car, one has a goat. It's a 50/50 chance, so it doesn't matter!"
>
> **The Mathematical Reality:** You should **always switch**. Switching doubles your chances of winning from 1/3 to 2/3.

2.18.3 Breaking Down the Probabilities

When you make your initial choice, there is a 1/3 chance you picked the car, and a 2/3 chance you picked a goat. Because Monty *must* reveal a goat, his action gives you new information.

If you originally picked...	Monty reveals...	Result if you STAY	Result if you SWITCH
Goat 1 (1/3 chance)	Goat 2	You get Goat 1	**You get the Car!**
Goat 2 (1/3 chance)	Goat 1	You get Goat 2	**You get the Car!**
The Car (1/3 chance)	A Goat	**You get the Car!**	You get a Goat

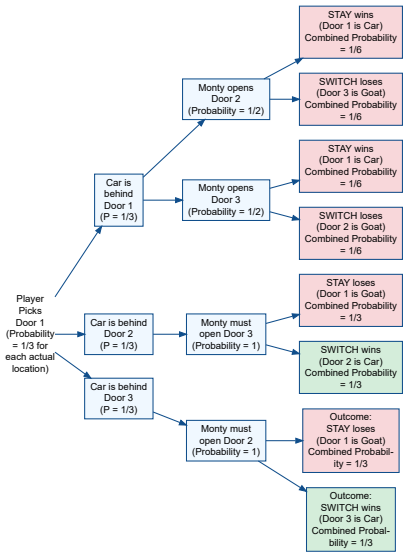
Notice that in 2 out of the 3 possible scenarios, switching wins you the car!

2.18.4 Have a go at the Game

2.18.5 Using Tree Diagram

To visualize why the odds shift, let's look at a probability tree. We will assume you start by picking **Door 1**.

Because Monty Hall's actions are restricted by the rules of the game (he *must* reveal a goat, and he *cannot* reveal your door), the tree branches unevenly.


Player Picks Door 1 (Probability = 1/3 for each actual location)
Car is behind Door 1 (P = 1/3)
Car is behind Door 2 (P = 1/3)
Car is behind Door 3 (P = 1/3)
Monty opens Door 2 (Probability = 1/2)
Monty opens Door 3 (Probability = 1/2)
Monty must open Door 3 (Probability = 1)
Monty must open Door 2 (Probability = 1)
STAY wins (Door 1 is Car) Combined Probability = 1/6
SWITCH loses (Door 3 is Goat) Combined Probability = 1/6
STAY wins (Door 1 is Car) Combined Probability = 1/6
SWITCH loses (Door 2 is Goat) Combined Probability = 1/6
STAY loses (Door 1 is Goat) Combined Probability = 1/3
SWITCH wins (Door 2 is Car) Combined Probability = 1/3
Outcome: STAY loses (Door 1 is Goat) Combined Probability = 1/3
Outcome: SWITCH wins (Door 3 is Car) Combined Probability = 1/3

2.18.6 Prove it in R

Don't believe the math? Let's simulate the game 10,000 times in R to see the Law of Large Numbers in action.

```r
# Set seed for reproducibility
set.seed(42)

# Number of games to play
n_games <- 10000

# Simulate the winning doors and the player's initial picks
winning_doors <- sample(1:3, n_games, replace = TRUE)
player_picks <- sample(1:3, n_games, replace = TRUE)

# If you STAY, you win only if your initial pick was the winning door
win_by_staying <- sum(player_picks == winning_doors)

# If you SWITCH, you win every time your initial pick was WRONG
# (because Monty removes the other wrong option)
win_by_switching <- sum(player_picks != winning_doors)

# Calculate win percentages
pct_stay <- (win_by_staying / n_games) * 100
pct_switch <- (win_by_switching / n_games) * 100

cat(sprintf("Win rate by staying: %.1f%%\n", pct_stay))
```

```
## Win rate by staying: 34.5%
```

```r
cat(sprintf("Win rate by switching: %.1f%%\n", pct_switch))
```

```
## Win rate by switching: 65.5%
```

Chapter 3

Chapter 2: Random Variables, Expectation, and Moments

3.1 Random Variables

- When analysing the result of an experiment, it is easier to deal with the *function* of the possible outcomes rather than the outcome itself
- For example, in an experiment of tossing a coin twice and observing the number of Heads, the possible outcomes are: TT, TH, HT, HH

  The function is **SUM** of Heads which are 0, 1, and 2.
- These are quantitiesa are **Random Variable**

Definition 3.1 (Random Variable)**.** Given a random experiment with an outcome space S, a function X that assigns one and only one real number $X(s) = x$ to each element s in S is called a **random variable**.

The **space** of X is the set of real numbers $\{x : X(s) = x, s \in S\}$, where $s \in S$ means that the element s belongs to the set S.

- A random variable, usually written X, is a variable whose possible values are numerical outcomes of a random phenomenon.
- A random variable assigns a numerical value to each outcome in a sample space.
- Looking it in mathematical function

a. Domain: TT, HT, TH, HH

b. Function, X: The rule you define (i.e. The number of Head)

c. Space of X, Range: 0, 1, 2

Example 3.1. A rat is selected at random from a cage and its sex is determined.

- The set of possible outcomes is female and male.
- The outcome space is $S = \{\text{female}, \text{male}\} = \{F, M\}$.
- Let X be a function defined on S such that $X(F) = 0$ and $X(M) = 1$.

- X is then a real-valued function that has the outcome space S as its domain and the set of real numbers $\{x : x = 0, 1\}$ as its range.
- We call X a random variable and the space associated with X is $\{x : x = 0, 1\}$.
- It may be possible that the set S has elements that are themselves real numbers.

-In such an instance, we could write $X(s) = s$ so that X is the identity function, and the space of X is also S.

Example 3.2. The experiment is tossing a die. Then the outcome space associated with this experiment is $S = \{1, 2, 3, 4, 5, 6\}$ with elements of S indicating the number of spots on the side facing up.

- For each $s \in S$, let $X(s) = s$.
- The space of the random variable X is then $\{1, 2, 3, 4, 5, 6\}$.

Exercise 3.1. Complete the table with space of X

Experiment	Possible Outcomes, S	Function X	Space of X
Toss two dice	(1, 1), (1, 2) … (6, 6)	X: Sum of the two dice	
Toss a coin 5 times	(TTTTT), (TTTTH), …., (HHHHH)	X: Number of Tail	
Apply different amount of fertilizer to palm trees in 2 acres	Weight of fertilizers	X: Weight of palm fruits / 2 acres	

Example 3.3. Tossing 3 fair coins. Y denoted the number of heads that appear. Then, Y is a RV taking on one of the values 0, 1, 2 and 3 with respective probabilities:

- $P(Y = 0) = P(TTT) = 1/8$
- $P(Y = 1) = P[(HTT, THT, TTH)] = 3/8$
- $P(Y = 2) = P[(HHT, HTH, THH)] = 3/8$
- $P(Y = 3) = P(HHH) = 1/8$

3.2 RV and Distribution Function

Random variable can be divided into

- Discrete
- Continuous

3.3 Discrete Random Variable

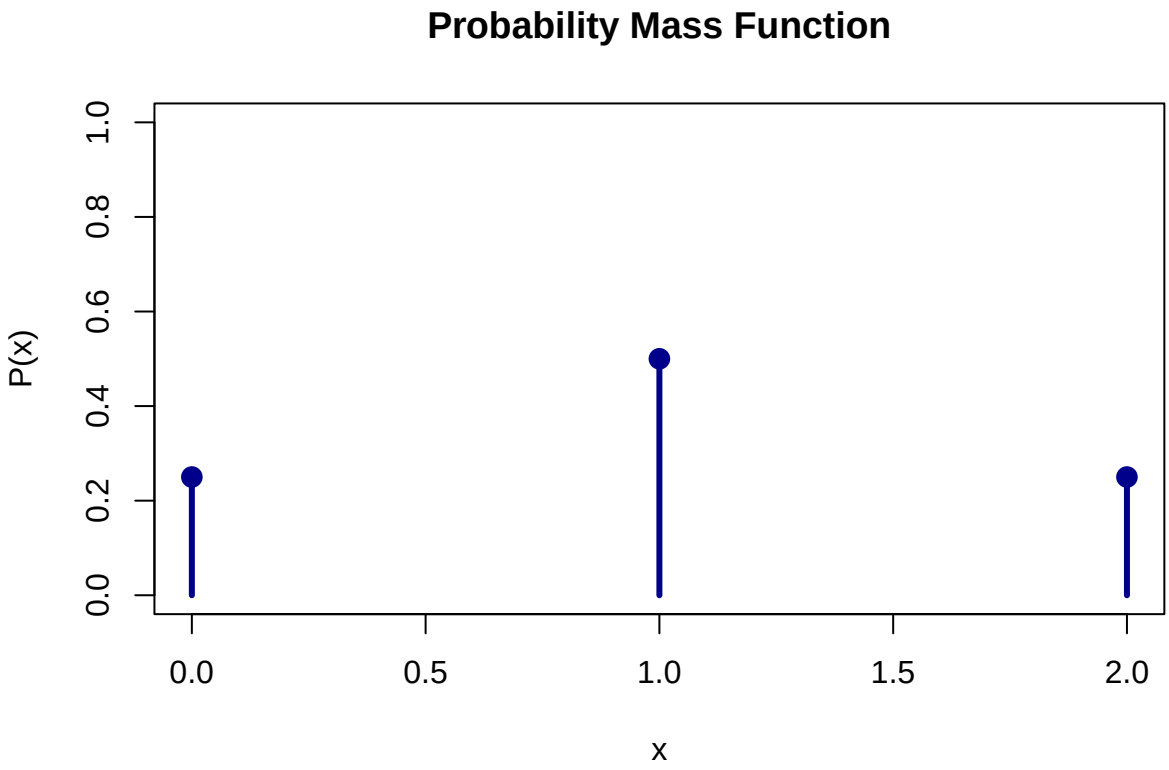
It is a random variable (RV) that can take on at most countable number of possible values.

Probability distribution for a discrete random variable:

- can be represented by a formula, a table, or a graph, which provides $P(X) = P(X = x)$ for all x.
- It assigns nonzero probabilities to only a countable number of distinct x values.
- Any value x not explicitly assigned a positive probability is understood to be such that $P(X = x) = 0$.
- The function $f(x) = p(x) = P(X = x)$ for each x within the range of X is called the probability distribution of X.
- It is often called the **probability mass function (p.m.f)** for the discrete random variable X.
- For the p.m.f, $f(x)$ is positive for at most a countable number of values of x and satisfies the following properties:
  1. $f(x_i) \geq 0, x \in S$
  2. $\sum_{x \in S} f(x_i) = 1$
  3. $P(x \in A) = \sum_{x \in A} P(x), A \subset S$

Example 3.4. We can plot the p.m.f in graphical format

x	0	1	2
$P(x)$	1/4	1/2	1/4

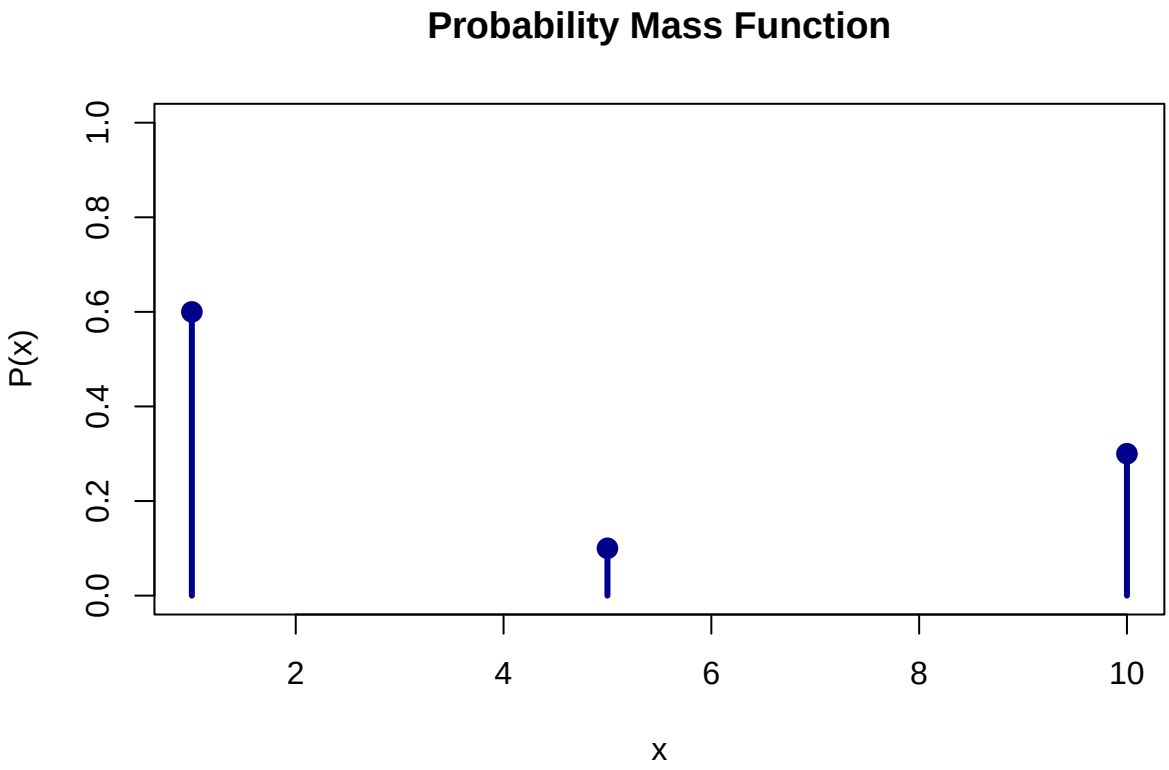


Exercise 3.2. Let a chip be taken at random from a bowl that contains six white chips, three red chips, and one blue chip. Let the random variable $X = 1$ if the outcome is a white chip, let $X = 5$ if the outcome is a red chip, and let $X = 10$ if the outcome is a blue chip.

a. Find the pmf of X.

x	1	5	10
$P(X = x)$	$\frac{3}{5}$	$\frac{1}{10}$	$\frac{3}{10}$

b. Graph the pmf as a line graph.



Exercise 3.3. A tech company manufactures high-end smartphones. Before shipping, an automated inspector checks batches of 4 phones for screen defects. Let X be the discrete random variable representing the number of defective phones found in a randomly selected batch of 4.Based on historical data, the probability mass function (PMF) $P(X = x)$ is given as follows:

x (Defects)	0	1	2	3	4
P(X=x)$	0.70	0.18	0.08	k	0.01

1. Find the value of k to complete the PMF.

Let X be the number of defect

$$\sum_{i=0}^{4} P(x_i) = 1 \tag{3.1}$$

$$1 - (0.7 + 0.18 + 0.08 + 0.01) = 0.03 \tag{3.2}$$

2. What is the probability that a batch has at least 2 defective phones?

$$P(X \geq 2) = P(X = 2) + P(X = 3) + P(X = 4) \tag{3.3}$$

$$= 0.08 + 0.03 + 0.01 = 0.12 \tag{3.4}$$

3. What is the probability that a batch has fewer than 3 defective phones?

$$P(X < 3) = 1 - P(X = 3) - P(X = 4) \tag{3.5}$$

$$= 1 - 0.03 - 0.01 = 0.96 \tag{3.6}$$

Exercise 3.4. A ride-hailing driver operates in a busy downtown district. Let Y be the discrete random variable representing the number of passenger ride requests the driver receives during a 15-minute period.

The PMF of Y is estimated as:

$$P(Y = y) = \begin{cases} 0.15 & \text{if } y = 0 \\ 0.40 & \text{if } y = 1 \\ 0.30 & \text{if } y = 2 \\ 0.15 & \text{if } y = 3 \\ 0 & \text{otherwise} \end{cases}$$

Verify that this function satisfies the two fundamental properties of a valid PMF.

- Non-negativity: All probabilities for each values that Y can take are all greater than 0
- Sum of the probabiility is equal to 1

$$0.15 + 0.40 + 0.30 + 0.15 + 0 = 1$$

3.4 Continuous Random Variable

X is a continuous RV if there exists a nonnegative function $f(x)$ defined for all real $x \in (-\infty, \infty)$, having the property that for any set B of real numbers, $P(X \in B) = \int_B f(x)dx$.

Probability distribution for a continuous random variable: $f(x)$ is approximately equal to the relative frequency of the interval $\{x : a < x < b\}$ and hence can be thought of as an estimate of the probability that the RV X falls into the interval (a, b). This probability is often written as $P(a < x < b) = \int_a^b f(x)dx$. It is often called the **probability density function (pdf)** for the continuous random variable X.

3.4.1 Properties of continuous Random Variable

The p.d.f. $f(x)$ satisfies:

- $f(x) > 0$ for $x \in S$.
- $\int_{-\infty}^{\infty} f(x)dx = 1$.
- $P(a < x < b) = \int_a^b f(x)dx$.
- Note: If $a = b$, then $P(x = a) = \int_a^a f(x)dx = 0$.

If $a, b \subseteq S$, then the probability of the event $\{a < x < b\}$ is $P(a < x < b) = \int_a^b f(x)dx$.

Remarks: if $a = b$, then $P(x = a) = \int_a^a f(x)dx = 0$.

Example 3.5. Suppose that X is a continuous random variable whose probability density function is given by:

$$f(x) = \begin{cases} C(4x - 2x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases} \tag{3.7}$$

a. What is the value of C?

b. Find $P(X > 1)$.

Solution:

a. Since f is a probability density function, we must have

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$C\int_0^2 (4x - 2x^2)dx = 1 \tag{3.8}$$

$$C\left[2x^2 - \frac{2x^3}{3}\right]_0^2 = 1 \tag{3.9}$$

$$C = \frac{3}{8} \tag{3.10}$$

b. Hence, $P(X > 1) = \frac{3}{8}\int_1^2 (4x - 2x^2)dx = \frac{1}{2}$.

Exercise 3.5. The amount of time in hours that a computer functions before breaking down is a continuous random variable with probability density function given by

$$f(x) = \begin{cases} \lambda e^{-x/100} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

What is the probability that:

a. A computer will function between 50 and 150 hours before breaking down?

b. It will function for fewer than 100 hours?

3.4.2 Cumulative Distributive Function (CDF)

Definition 3.2 (Cumulative Distributive Function (CDF))**.** The cumulative distribution function or CDF of a RV X, denoted by $F(x)$ is defined by

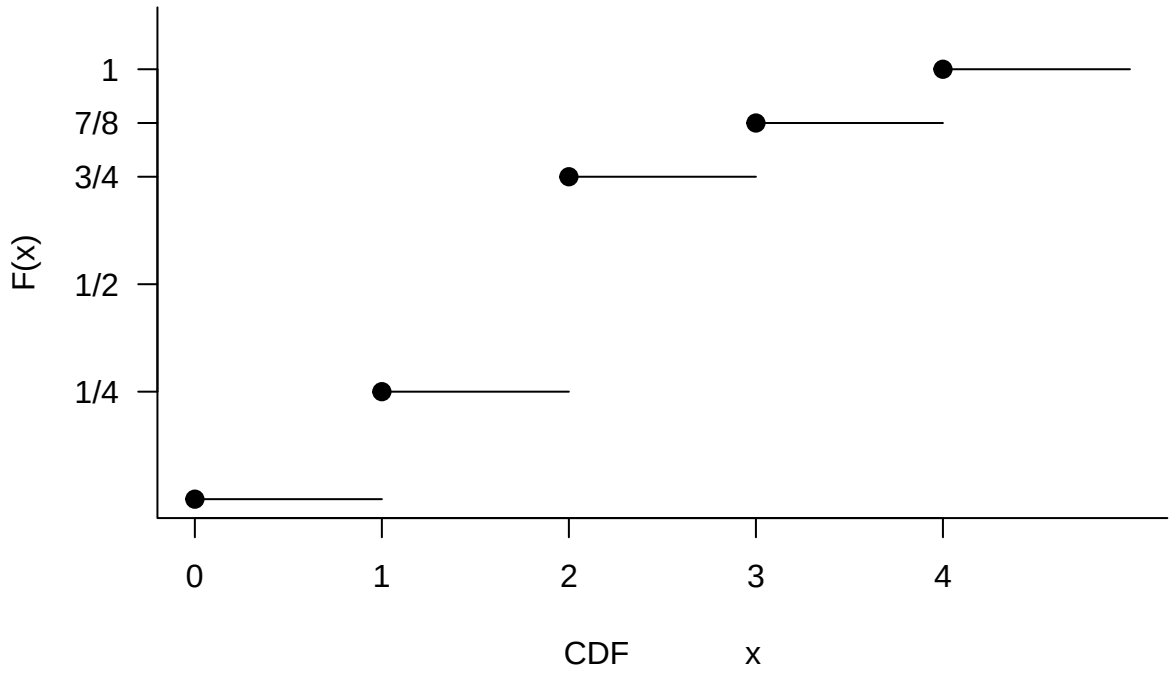
$$F(x) = P(X \leq x), \text{ for all } x.$$

$$F(x) = \begin{cases} \sum_{X \leq x} P(x) & \text{discrete R.V} \\ \int_{-\infty}^{x} f(t)dt & \text{continuous R.V} \end{cases}$$

Example 3.6. Consider the table below:

x	1	2	3	4
$P(x)$	1/4	1/2	1/8	1/8

$$F(x) = \begin{cases} 0 & x < 1 \\ 1/4 & 1 \leq x < 2 \\ 3/4 & 2 \leq x < 3 \\ 7/8 & 3 \leq x < 4 \\ 1 & 4 \leq x \end{cases}$$



Example 3.7. Let X has the p.d.f of $f(x) = \begin{cases} 0 & -\infty < x < 0 \\ \frac{1}{20}e^{-x/20} & 0 \leq x < \infty \end{cases}$

To obtain C.D.F,

$$F(x) = \int_{-\infty}^{x} f(t)dt = \int_{0}^{x} \frac{1}{20} e^{-t/20} dt = \left[-e^{-t/20}\right]_{0}^{x} = 1 - e^{-x/20}$$

Also, $F'(x) = f(x)$.

3.4.3 Relationship between CDF, $F(x)$ and p.m.f or p.d.f

$F(x) \rightarrow$ p.m.f , p.d.f

- If discrete, $P(x = a) = F(a) - F(a^{-})$
- If continuous, $f(x) = \frac{dF(x)}{dx} = F'(x)$

p.m.f, p.d.f $\rightarrow F(x)$

- If discrete, $F(x) = \sum P(x)$
- If continuous, $F(x) = \int_{-\infty}^{x} f(x)dx$

Transformation	Variable Type	Formula
F(x) → p.m.f, p.d.f	If discrete	$P(x = a) = F(a) - F(a^{-})$
	If continuous	$f(x) = \frac{dF(x)}{dx} = F'(x)$
p.m.f, p.d.f → F(x)	If discrete	$F(x) = \sum P(x)$
	If continuous	$F(x) = \int_{-\infty}^{x} f(x)dx$

3.4.4 Properties of CDF

$F(x)$ is CDF if and only if the following 3 conditions hold:

a. $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$

b. $F(x)$ is non decreasing function of x

c. $F(x)$ is right continuous

Example 3.8. A batch of 15 items contains 6 defective ones. A random sample of 5 is chosen. Let X be the number of defective items in the sample. Find the probability function and cumulative distribution function for X.

Solution

Exercise 3.6. Let $X \in \mathbb{R}$ with PDF as

$$f(x) = \begin{cases} 0 & -\infty < x < 0 \\ \frac{1}{20} \exp\left(-\frac{x}{20}\right) & 0 \leq < \infty \end{cases} \tag{3.11}$$

Derive the CDF of X.

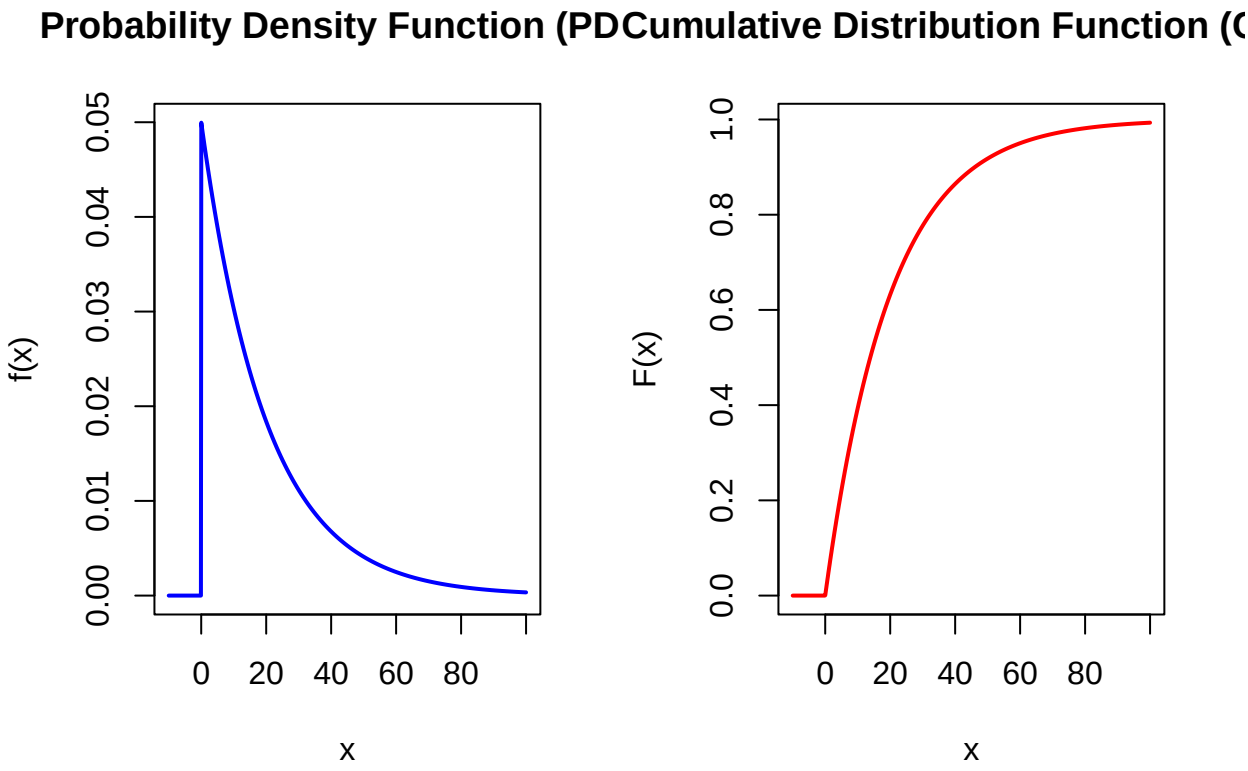
Solution

$$F(x) = \int_{-\infty}^{0} f(t)\, dt \tag{3.12}$$

$$= \int_{0}^{x} \frac{1}{20} \exp\left(-\frac{x}{20}\right)\, dt \tag{3.13}$$

$$= \left[-\frac{x}{20}\right]_{0}^{x} = 1 - \exp\left(-\frac{x}{2-}\right) \tag{3.14}$$

Check: the derivative of $F(x)$ must equal to $f(x)$



Exercise 3.7. The random variable X has probability density function given by

$$f(x) = \begin{cases} \frac{k}{(1+x)^4} & \text{for } x \geq 0 \\ 0 & \text{for } x \leq 0 \end{cases} \tag{3.15}$$

For some positive constant k. Find the value of k, the distribution function $F(x)$ and the probabilities $P(X \leq 1)$, $P(X > 1)$ and $P(1 \leq X \leq 2)$.

Solution:

1. Find the value of k For $f(x)$ to be a valid probability density function, the total area under the curve

must equal 1:

$$\int_{-\infty}^{\infty} f(x)dx = 1 \tag{3.16}$$

$$\int_{0}^{\infty} \frac{k}{(1+x)^4}dx = 1 \tag{3.17}$$

To integrate, we use the power rule. Let $u = 1 + x$, then $du = dx$:

$$k\left[\frac{(1+x)^{-3}}{-3}\right]_0^{\infty} = 1 \tag{3.18}$$

$$-\frac{k}{3}\left(\lim_{x\to\infty}\frac{1}{(1+x)^3} - \frac{1}{(1+0)^3}\right) = 1 \tag{3.19}$$

$$-\frac{k}{3}(0-1) = 1 \tag{3.20}$$

$$\frac{k}{3} = 1 \implies k = 3 \tag{3.21}$$

$$\tag{3.22}$$

Thus, the PDF is $f(x) = \frac{3}{(1+x)^4}$ for $x \geq 0$.

2. Find the cumulative distribution function, $F(x)$ By definition, $F(x) = \int_{-\infty}^{x} f(t)dt$.

For $x \geq 0$:

$$F(x) = \int_0^x \frac{3}{(1+t)^4}dt \tag{3.23}$$

$$= 3\left[\frac{(1+t)^{-3}}{-3}\right]_0^x \tag{3.24}$$

$$= -1\left(\frac{1}{(1+x)^3} - \frac{1}{(1+0)^3}\right) \tag{3.25}$$

$$= 1 - \frac{1}{(1+x)^3} \tag{3.26}$$

The complete distribution function is:

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 - \frac{1}{(1+x)^3} & \text{for } x \geq 0 \end{cases}$$

3. Calculate the Probabilities Using the CDF $F(x)$ we just derived:

Find $P(X \leq 1)$:

$$P(X \leq 1) = F(1) = 1 - \frac{1}{(1+1)^3} \tag{3.27}$$

$$= 1 - \frac{1}{8} = \frac{7}{8} = 0.875 \tag{3.28}$$

Find $P(X > 1)$**:**

$$P(X > 1) = 1 - P(X \leq 1) \tag{3.29}$$
$$= 1 - \frac{7}{8} = \frac{1}{8} = 0.125 \tag{3.30}$$

Find $P(1 \leq X \leq 2)$**:**

$$P(1 \leq X \leq 2) = F(2) - F(1) \tag{3.31}$$
$$F(2) = 1 - \frac{1}{(1+2)^3} = 1 - \frac{1}{27} = \frac{26}{27} \tag{3.32}$$
$$P(1 \leq X \leq 2) = \frac{26}{27} - \frac{7}{8} = \frac{208}{216} - \frac{189}{216} = \frac{19}{216} \approx 0.08796 \tag{3.33}$$
$$\tag{3.34}$$

3.5 Expected Values and Expected Function

It is a value that were expected to emerge when performing some random experiment. So basically, it is average or mean of the outcomes.

Definition 3.3 (Expected Values and Expected Function)**.** The expected value or mean of a random variable X, denoted by $E(X)$ is:

$$\mu = E(X) = \begin{cases} \int_{-\infty}^{\infty} xf(x)dx & \text{if } X \text{ is continuous} \\ \sum xf(x) = \sum g(x)P(X = x) & \text{if } X \text{ is discrete} \end{cases}$$

Example 3.9 (Example 8)**.** Let X equal number of spot on the side facing upwards after a six-sided die (one of pair of dice) is cast at a random. If everything is fair about this experiment, a reasonable probability model is given by the P.M.F.

$$f(x) = P(X = x) = \frac{1}{6}$$

where $x = 1, \ldots, 6$.

Find the expected value of the distribution of X.

Solution

X is the number of spot on the side facing upwards after a six-sided die. To find the expected value,

$$E[X] = \sum xf(x) = \sum_{1}^{6} x\frac{1}{6} \tag{3.35}$$
$$= (1 + 2 + 3 + 4 + 5 + 6)\frac{1}{6} \tag{3.36}$$
$$= \frac{7}{2} = 3.5 \tag{3.37}$$

Example 3.10 (Example 9)**.** From previous Example 8, Let $Y = X^2$, then Y is also a random variable such that p.m.f of Y is $g(y) = P(Y = y) = 1/6$, where $y = 1, 4, 9, 16, 25, 36$. Suppose we wanted to find the expected value of Y, $E(Y)$, and $Y = X^2$, then we can also compute:

$$E(Y) = E(X^2) = \sum_{x=1}^{6} x^2 f(x) = \frac{1^2}{6} + \frac{2^2}{6} + \frac{3^2}{6} + \frac{4^2}{6} + \frac{5^2}{6} + \frac{6^2}{6} = \frac{91}{6}$$

Note that, $E(X^2)$ also known as second moment about the origin with respect to variable X.

Example 3.11 (Example 10)**.** Given that X is the number of spot on the side facing upwards after a six-sided die.

Let Y be the squared of number of spot on the side facing upwards after a six-sided die with the P.M.F of

$$g(y) = P(Y = y) = \frac{1}{6} \tag{3.38}$$

where $y = 1, 4, 9, 16, 25, 36$. Find the expectation of Y.

Solution

- In this question, Y is a function of X where $Y = X^2$
- The value of Y is no longer the value shown on the surface
- To find the expectation of Y, we must change the formula

$$E[Y] = \sum yf(y) = \sum_{1}^{6} y\frac{1}{6} \tag{3.39}$$

$$= (1 + 4 + 9 + 16 + 25 + 36)\frac{1}{6} \tag{3.40}$$

$$= \frac{91}{6} = 15.167(3\text{dp}) \tag{3.41}$$

- $E[Y] = E[X^2]$ is the **second moment about the origin** with respect to variable X

Exercise 3.8. A batch of 15 items contains 6 defective ones. A random sample of 5 is chosen. Let X be the number of defective items in the sample.

Calculate the expected value of the distribution of X.

Example 3.12 (Example 11)**.** Let $X \in \mathbb{R}$ be a random variable with PDF of

$$f(x) = \begin{cases} 2x & \text{when} \quad 0 \leq x \leq 1 \\ \\ 0 & \text{otherwise} \end{cases} \tag{3.42}$$

Calculate the expected value of the distribution of X.

Solution

$$E[X] = \int_0^1 xf(x) \tag{3.43}$$

$$= \int_0^1 2x^2 \tag{3.44}$$

$$= \left[\frac{2x^3}{3}\right]_0^1 \tag{3.45}$$

$$= \frac{2}{3} \tag{3.46}$$

3.5.1 Expected Values: Properties

Let $a, c \in \mathbb{R}$ and X and Y be random variables with the same sample space.

- Expectation of a constant is a constant: $E(c) = c$.
- $E[aX] = aE[X]$.
- Linearity: $E[X] + c = E[X] + c$.
- Additivity: $E[X + Y] = E[X] + E[Y]$.
- $|E(X)| \leq E(|X|)$.

3.6 Variance and standard deviation

Calculation for Variance and standard deviation can also be calculated using expectation. Where:

$$Var(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

$$\sigma = \sqrt{Var(X)}$$

The **variance**, denoted by σ^2.

- Continuous: $Var[X] = \int_{-\infty}^{\infty} x^2 f(x)dx - (\int_{-\infty}^{\infty} xf(x)\ dx)^2$
- Discrete: $Var[X] = \sum x^2 f(x) - (\sum xf(x))^2$
  - provided that the integral or sum exists
- SD, σ, is just the square root of variance

Example 3.13 (Example 16)**.** Let the P.M.F. of X be defined by $f(x) = x/6$, where $x = 1, 2, 3$.The mean of X is

$$\mu = E[X] = 1\left(\frac{1}{6}\right) + 2\left(\frac{2}{6}\right) + 3\left(\frac{3}{6}\right) = \frac{7}{3}$$

Find the variance and standard deviation of X.

Solution: Let $X = 1, 2, 3$ with the P.M.F of $f(x) = x/6$

$$Var[X] = E[(X-\mu)^2] = E[X^2] - \mu^2 \tag{3.47}$$
$$= E[X^2] - \frac{7}{3} \tag{3.48}$$

The second order moment of X is $E[X^2]$

$$E[X^2] = 2\left(\frac{1}{6}\right) + 4\left(\frac{2}{6}\right) + 9\left(\frac{3}{6}\right) = \frac{37}{6} = 6.167 \text{ (3 dp)} \tag{3.49}$$

Example 3.14 (Example 17)**.** Compute the variance of the discrete distribution where

- $p(0) = 35/210$
- $p(1) = 105/210$
- $p(2) = 63/210$ and
- $p(3) = 7/210$.

Solution:

First, calculate the expected value $E[X]$:

$$E[X] = \sum_{x=0}^{3} x \cdot p(x) \tag{3.50}$$
$$= (0)\left(\frac{35}{210}\right) + (1)\left(\frac{105}{210}\right) + (2)\left(\frac{63}{210}\right) + (3)\left(\frac{7}{210}\right) \tag{3.51}$$
$$= 0 + \frac{105}{210} + \frac{126}{210} + \frac{21}{210} = \frac{252}{210} = 1.2 \tag{3.52}$$

Next, calculate the second moment, $E[X^2]$:

$$E[X^2] = \sum_{x=0}^{3} x^2 \cdot p(x) \tag{3.53}$$
$$= (0^2)\left(\frac{35}{210}\right) + (1^2)\left(\frac{105}{210}\right) + (2^2)\left(\frac{63}{210}\right) + (3^2)\left(\frac{7}{210}\right) \tag{3.54}$$
$$= 0 + \frac{105}{210} + \frac{252}{210} + \frac{63}{210} = \frac{420}{210} = 2 \tag{3.55}$$

Finally, compute the variance $\text{Var}(X)$ using the variance formula:

$$\text{Var}(X) = E[X^2] - (E[X])^2 \tag{3.56}$$
$$= 2 - (1.2)^2 = 2 - 1.44 = 0.56 \tag{3.57}$$

3.6.1 Property of Variance

Let X and Y be random variable and $a, b \in \mathbb{R}$.

a. **Non-negative**: $Var[X] \geq 0$

b. **Constant has 0 variance**: $Var[a] = 0$

c. $Var[aX + b] = a^2 Var[X]$

d. $Var[aX + bY] = a^2 Var[X] + b^2 Var[Y]$

Example 3.15 (Example 18)**.** Show that $Var[a + bX] = b^2 Var[X]$.

Solution:

By the definition of variance, for any random variable Y, we have:

$$\text{Var}(Y) = E[(Y - E[Y])^2]$$

Let $Y = a + bX$. First, we determine the expected value of Y by applying the linearity of expectation:

$$E[a + bX] = a + bE[X]$$

Next, we substitute $a + bX$ and its expected value into the variance formula:

$$\begin{aligned} \text{Var}(a + bX) &= E[((a + bX) - E[a + bX])^2] && (3.58) \\ &= E[((a + bX) - (a + bE[X]))^2] && (3.59) \end{aligned}$$

Simplify the terms inside the inner parentheses by canceling out the constant a:

$$\begin{aligned} \text{Var}(a + bX) &= E[(bX - bE[X])^2] && (3.60) \\ &= E[(b(X - E[X]))^2] && (3.61) \\ &= E[b^2(X - E[X])^2] && (3.62) \end{aligned}$$

Because b^2 is a constant, it can be factored out of the expectation:

$$\text{Var}(a + bX) = b^2 E[(X - E[X])^2]$$

Finally, by recognizing that $E[(X - E[X])^2]$ is the formal definition of $\text{Var}(X)$, we conclude the proof:

$$\text{Var}(a + bX) = b^2 \text{Var}(X)$$

Example 3.16 (Example 19)**.** Let X be a contious random variable with the P.D.F

$$f(x) = \begin{cases} 2x & for \quad 0 \leq x \leq 1 \\ 0 & for \quad \text{otherwise} \end{cases} \tag{3.63}$$

Find the expected value, standard deviation of X.

Solution

$$E[X] = \int xf(x) \tag{3.64}$$

$$= \int_0^1 x(2x) = \int_1^1 2x^2 \tag{3.65}$$

$$= \left[\frac{2}{3}x^3\right]_0^1 = \frac{2}{3} \tag{3.66}$$

$$\tag{3.67}$$

3.7 Functions of Random Variable

If X is a random variable with CDF $F_X(x)$, the function of X (e.g. $g(X)$) is also a random variable.

- We can rewrite the $g(X)$ as $Y = g(X)$
- Hence, Y is a function of X

3.7.1 Functions of Random Variable: Expected Values

If X is a random variable and $g(X)$ is a function of X. The expected value of $E[g(X)]$ is

$$E[g(X)] = \begin{cases} \int g(x)f_X(x)\ dx & \text{if } X \text{ is continuous} \\ \\ \sum g(x)f_X(x) = \sum g(x)P(X = x) & \text{if } X \text{ is discrete} \end{cases} \tag{3.68}$$

Example 3.17 (Example 13 (Continuous case))**.** Let X be a continuous random variable representing the lifetime of a specific machine component (in years), with the P.D.F:

$$f(x) = \begin{cases} \frac{1}{2}\exp\left(-\frac{x}{2}\right) & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases} \tag{3.69}$$

Suppose the maintenance cost (in RM) over the component's lifetime is given by the function $g(X) = 50 + 10X^2$.

Find the expected maintenance cost, $E[g(X)]$.

Solution: By the linearity of expectation, we know that:

$$E[g(X)] = E[50 + 10X^2] = 50 + 10E[X^2]$$

First, find the second moment, $E[X^2]$:

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x)dx \tag{3.70}$$

$$= \int_0^{\infty} x^2 \left(\frac{1}{2}\exp\left(-\frac{x}{2}\right)\right) dx \tag{3.71}$$

Using integration by parts (or the Gamma function property $\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$):

$$E[X^2] = \frac{1}{2}\left(\frac{2!}{(1/2)^3}\right) \tag{3.72}$$

$$= \frac{1}{2}(16) = 8 \tag{3.73}$$

Now, substitute $E[X^2]$ back into the cost function:

$$E[g(X)] = 50 + 10(8) \tag{3.74}$$

$$= 50 + 80 = 130 \tag{3.75}$$

The expected maintenance cost is **RM 130**.

3.7.2 Functions of Random Variable: Expected Values Properties

Let X be a random variable and $a, b, c \in \mathbb{R}$. Then, for any functions $g_1(x)$ and $g_2(x)$ whose expectations exists:

a. $E[ag_1(x) + bg_2(x) + c] = aEg_1(x) + bE[g_2(x)] + c$

b. If $g_1(x) \geq 0$ for all x, then $E[g_1(x)] \geq 0$

c. If $g_1(x) \geq g_2(x)$ for all x, then $E[g_1(x)] \geq E[g_2(x)]$

d. If $a \leq g_1(x) \leq b$ for all x, then $a \leq E[g_1(x)] \leq b$

Example 3.18 (Example 14)**.** Suppose Grab charges a fixed fare of RM4 plus a mileage charge of RM2 per kilometer. Write the

If the journey length in kilometers is regarded as a random variable X, the linearity property

$$E[4 + 2X] = 4 + 2E[X]$$

just says that the average journey fare is RM4 plus RM2 times the average journey length. This is true whatever the distribution of X is: the fare depends linearly on the journey length.

Example 3.19 (Example 15)**.** Uber charges a fare in RM of $y(X) = 4\sqrt{1+X}$ for a journey of length X kilometers. Suppose the probability density function for X is

$$f(x) = \frac{3}{(1+x)^4}$$

.

Find the mean fare.

Solution:

To find the expected fare, we need to calculate the expected value of the fare function $y(X) = 4\sqrt{1+X}$, given the continuous probability density function

$$f(x) = \frac{3}{(1+x)^4}$$

for journey lengths $x \geq 0$

Using the definition of expectation for a function of a continuous random variable, we integrate over the support of X:

$$E[y(X)] = \int_0^\infty y(x)f(x)dx$$

Substitute the given equations into the integral:

$$E[y(X)] = \int_0^\infty \left(4\sqrt{1+x}\right)\left(\frac{3}{(1+x)^4}\right)dx$$

Combine the terms by converting the square root to a fractional exponent:

$$E[y(X)] = \int_0^\infty 12(1+x)^{1/2}(1+x)^{-4}dx$$

$$E[y(X)] = 12\int_0^\infty (1+x)^{-7/2}dx$$

Perform the integration:

$$E[y(X)] = 12\left[\frac{(1+x)^{-5/2}}{-5/2}\right]_0^\infty \tag{3.76}$$

$$= 12\left(-\frac{2}{5}\right)\left[(1+x)^{-5/2}\right]_0^\infty \tag{3.77}$$

$$= -\frac{24}{5}\left(\lim_{x\to\infty}\frac{1}{(1+x)^{5/2}} - \frac{1}{(1+0)^{5/2}}\right) \tag{3.78}$$

$$= -\frac{24}{5}(0-1) \tag{3.79}$$

$$= \frac{24}{5} \tag{3.80}$$

$$= 4.8 \tag{3.81}$$

The expected fare for the journey is **RM 4.80**.

Exercise 3.9 (Example 19)**.** Find $E(X)$, $Var(X)$ and Standard deviation of X when the density function of X is $f(x) = 2x$ if $0 \le x \le 1$, and 0 if x is otherwise.

Solution:

Given the continuous probability density function $f(x) = 2x$ for $0 \le x \le 1$[cite: 11].

1. Calculate the Expected Value, $E(X)$: Use the formula for the continuous expected value:

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx \tag{3.82}$$

$$= \int_0^1 x(2x)dx \tag{3.83}$$

$$= \left[\frac{2x^3}{3}\right]_0^1 \tag{3.84}$$

$$= \frac{2(1)^3}{3} - 0 = \frac{2}{3} \tag{3.85}$$

$$\tag{3.86}$$

2. Calculate the Second Moment, $E[X^2]$:

$$E(X^2) = \int_0^1 x^2 f(x)dx \tag{3.87}$$
$$= \int_0^1 x^2(2x)dx \tag{3.88}$$
$$= \int_0^1 2x^3 dx \tag{3.89}$$
$$= \left[\frac{2x^4}{4}\right]_0^1 \tag{3.90}$$
$$= \frac{2}{4} = \frac{1}{2} \tag{3.91}$$

3. Calculate the Variance, $Var(X)$:

$$Var(X) = E(X^2) - (E(X))^2 \tag{3.92}$$
$$= \frac{1}{2} - \left(\frac{2}{3}\right)^2 \tag{3.93}$$
$$= \frac{1}{2} - \frac{4}{9} \tag{3.94}$$
$$= \frac{9}{18} - \frac{8}{18} = \frac{1}{18} \tag{3.95}$$
$$\tag{3.96}$$

4. Calculate the Standard Deviation:

$$\text{SD}[X] = \sqrt{Var(X)} \tag{3.97}$$
$$= \sqrt{\frac{1}{18}} = \frac{1}{\sqrt{18}} \tag{3.98}$$
$$\approx 0.2357 \tag{3.99}$$

3.7.3 Expectation of function

- If X is a discrete RV, with $x_i, i \geq 1$ with probabilities $p(x_i)$, then $E[g(X)] = \sum_i g(x_i)p(x_i)$.
- Let X be a random variable with density $f(x)$. The expected value of a function $g(\cdot)$ of the random variable, denoted $E(g(X))$, is defined to be $E(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)dx$.

3.7.4 Expectation of function Properties

Let X be a RV and a, b and c are constants. Then, for any functions $g_1(x)$ and $g_2(x)$ whose expectations exist:

a. $E(ag_1(x) + bg_2(x) + c) = aEg_1(x) + bEg_2(x) + c$

b. If $g_1(x) \geq 0$ for all x, then $Eg_1(x) \geq 0$

c. If $g_1(x) \geq g_2(x)$ for all x, then $Eg_1(x) \geq Eg_2(x)$

d. If $a \leq g_1(x) \leq b$ for all x, then $a \leq Eg_1(x) \leq b$

3.8 Moments and Moment generating function

3.8.1 Moments

Definition 3.4 (Methods of Moments)**.** The moments of RV or of a distributions are the expectations of the powers of the RV which has the given distributions. If X is a RV, the rth moment of x is defined as $\mu_r' = E[X^r]$. Therefore, the rth moment about the origin of a random variable is the expected value of X^r:

$$\mu_r^{'} = E[X^r] = \begin{cases} \sum_x x^r f(x) \\ \\ \int x^r f(x) \end{cases} \tag{3.100}$$

- The first moment is the mean, i.e. $E[X]$

The first moment is $E(X)$.

3.8.2 Central Moments

Definition 3.5 (Centrol Moments)**.** The rth moment about the mean of a random variable X, is the expected value of $(X - \mu_X)^r$, symbolically,

$$\mu_r = E[(X - \mu_X)^r] = \begin{cases} \sum_x (x - \mu_X)^r f(x) \\ \int_{-\infty}^{\infty} (x - \mu_X)^r f(x) dx \end{cases}$$

The rth moment about the mean of a random variable X is sometimes called the rth central moment of X. Note that: * $\mu_1 = E[(X - \mu_X)] = 0$ * $\mu_2 = E[(X - \mu_X)^2] = Var[X]$ Variance can be also written as $\sigma_X^2 = \mu_2' - \mu_X^2$.

The first moment is mean of $E(X)$. The second central moment is the variance. If $E(X^n)$ exists for some +ve integers n, we call $E(X^n)$ the nth moment of X about the origin. If $E|X^n| < \infty$ for some +ve real number α, we call $E|X^n|$ the αth absolute moment of X.

3.8.3 Moment Generating Function (MGF)

Definition 3.6 (Moment Generating Function (MGF))**.** Moment generating function $M(t)$ of the RV X is defined for all real values of t by:

$$M(t) = E[e^{tx}] = \begin{cases} \sum e^{tx} p(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} e^{tx} \cdot f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Example 3.20 (Example 12 (Discrete case))**.** Let the p.m.f. of X be defined by $f(x) = x/6$, where $x = 1, 2, 3$. Find the mgf of RV of X.

Example 3.21 (Example 21 (Continuous case))**.** X is Random variable with probability density function of X is

$$f(x) = \begin{cases} 1/6 & \text{for } 1 \leq x \leq 7 \\ 0 & \text{otherwise} \end{cases} \tag{3.101}$$

Find the mgf of X.

3.8.4 Finding probability distribution from mgf

Sometimes we can find the probability distribution of the random variable from observing their mgf. Since mgf is unique to each RV if the mgf exists. For discrete case; let space of S is $\{b_1, b_2, b_3, ...\}$, then the moment-generating function is given by the expansion:

$$M_X(t) = e^{tb_1} f(b_1) + e^{tb_2} f(b_2) + e^{tb_3} f(b_3) + ...$$

Thus, the coefficient of e^{tb_i} is the probability of $f(b_i) = P(X = b_i)$.

For continuous case; it is not as easy as the discrete case since it does not involve with expansion. But since mgf is unique to one random variable, we can see that in later topic, it is easy to find the pdf of given mgf.

3.8.5 Properties of mgf

$$M'(t) = \frac{d}{dt}E[e^{tx}] = E\left[\frac{d}{dt}(e^{tx})\right] = E[Xe^{tx}]$$

$$M'(0) = E(x)$$

$$M''(t) = \frac{d}{dt}M'(t) = \frac{d}{dt}E[Xe^{tx}] = E\left[\frac{d}{dt}(Xe^{tx})\right] = E[X^2e^{tx}]$$

$$M''(0) = E(X^2)$$

1. If $M_X(t)$ exists, then for any positive integer k, $\frac{d^k M_X(t)}{dt^k}\Big|_{t=0} = M_X^{(k)}(0) = E(X^k)$, i.e. we can find the kth mgf through the kth derivative of with respect to t by setting $t = 0$ (discrete case). We call $M_X(t)$ the mgf because all the moments of X can be obtained by differentiating $M_X(t)$ and then evaluating the result at $t = 0$.

Example 3.22 (Example 11 (Cntinuous case))**.** using the same problem in example 11, find mean and variance of X using the mgf.

If X has mgf, then $E(X^k) = \frac{d^k M_X(t)}{dt^k}\Big|_{t=0} = M_X^{(k)}(0)$, i.e. the nth moment of the distribution is equal to the nth derivative evaluated at $t = 0$ (continuous case).

For independent variable X and Y; $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$ and for any constant $a, b \in \mathbb{R}$: * $M_{X+a}(t) = E(e^{(X+a)t}) = e^{at} \cdot M_X(t)$ * $M_{bX}(t) = E(e^{bXt}) = M_X(bt)$ * $M_{\frac{X+a}{b}}(t) = E\left(e^{\frac{X+a}{b}t}\right) = e^{\frac{a}{b}t} M_X\left(\frac{t}{b}\right)$

Exercise 3.10. If the moment-generating function of X is

$$M(t) = \frac{2}{5}e^t + \frac{1}{5}e^{2t} + \frac{2}{5}e^{3t}$$

Find the mean, variance, and pmf of X.

3.9 Tutorials

1. Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed 6 times. What are the possible values of X?

2. Computer chips often contain surface imperfections. For a certain type of computer chip, the probability mass function of the number of defects X is presented in the following table.

x	0	1	2	3	4
$P(X = x)$	0.4	0.20	0.35	0.15	0.10

Answer the following:

a. Find $P(X \leq 3)$.
b. Find $P(X > 2)$.
c. Find μ_X .
d. Find σ^2
e. Find σ

3. A survey of cars on a certain stretch of highway during morning commute hours showed that 50% had only one occupant, 25% had 2, 20% had 3, 3% had 4, and 2% had 5. Let X represent the number of occupants in a randomly chosen car.

a. Find $P(X \leq 3)$.
b. Find $P(X > 2)$.
c. Find μ_X .
d. Find σ^2
e. Find σ

4. The delay time (in minutes) for vehicles passing through the Mines Highway (Besraya) toll plaza during morning peak hours is modeled by a continuous random variable T.

The probability density function (p.d.f.) for the delay time is given by:

$$f(t) = \begin{cases} \frac{1}{2} - \frac{1}{8}t & 0 \leq t \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

a. Find $P(T \leq 2)$.b. Find $P(T > 3)$.c. Find the expected delay time, μ_T.d. Find the variance, σ^2.e. Find the standard deviation, σ.

5. The waiting time (in hours) for patients to see a medical officer at a public hospital outpatient clinic in Selangor is modeled by a continuous random variable X. The probability density function (p.d.f.) for the waiting time is given by:

$$f(x) = \begin{cases} \frac{2}{9}x & 0 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

Instead of finding the probabilities, the hospital management wants to determine specific target waiting times (x) based on their Key Performance Indicators (KPIs).

a. Find the waiting time x such that exactly 25% of the patients wait less than x hours (i.e., find x where $P(X \leq x) = 0.25$).
b. Find the waiting time x such that the probability a patient waits longer than x hours is 0.64 (i.e., find x where $P(X > x) = 0.64$).
c. Find the median waiting time x (the time where $P(X \leq x) = 0.50$).
d. Find the expected waiting time a patient will experience, $E(X)$.

e. Patients must pay a hospital parking fee of RM 4.00 per hour of waiting time, alongside a fixed clinic registration fee of RM 5.00. Let $Y = 4X + 5$ represent the total out-of-pocket cost incurred by a patient before seeing the doctor. Using the linear properties of expectation and variance, calculate the expected total cost, $E(Y)$, and the variance of the total cost, $Var(Y)$.

6. The number of customers in line at a supermarket express checkout counter is a random variable whose probability mass function is given in the following table.

x	0	1	2	3	4	5
$P(X = x)$	0.15	0.20	0.35	0.15	0.10	0.05

Let X denote the number of customers in line, and let Y denote the total number of items purchased by all the customers in line. Assume the number of items purchased by one customer is independent of the number of items purchased by any other customer.

a. Find $P(X = 2 \text{and} Y = 2)$.
b. Find $P(X = 2 \text{and} Y = 6)$.
c. Find $P(Y = 2)$.

7. A local meteorologist in Dengkil is analyzing daily weather patterns during the November monsoon season. The experiment consists of recording the primary weather condition for a randomly selected day.Based on historical data, the outcome space S consists of four weather conditions with the following probabilities:

Weather conditions	Sunny (S)	Cloudy (C)	Rainy (R)	$Thunderstorm (T)
Probability	0.15	$0.35	0.40	0.10

To evaluate the potential impact on local infrastructure, the meteorologist defines a discrete random variable X representing the "Weather Severity Index." The index assigns a value of:

- 0 for Sunny
- 1 for Cloudy
- 3 for Rainy
- 5 for a Thunderstorm.

Answer the following:

a. Construct the probability mass function (PMF) table for the random variable X. (4 marks)

b. Calculate the probability that a randomly selected day has a severity index of strictly less than 3. (2 mark)

c. Calculate the expected Weather Severity Index, $E(X)$, for a day in November. (3 marks)

Chapter 4

Random Variables and Distributions

Chapter 5

Overview

This is a continuous notes

Chapter 6

Random Walk for Discrete Random Variable

The mathematical concept of the "drunken walk" - formally known as the random walk—was developed independently at the turn of the 20th century by thinkers across finance, biology, and physics attempting to model unpredictable behavior.

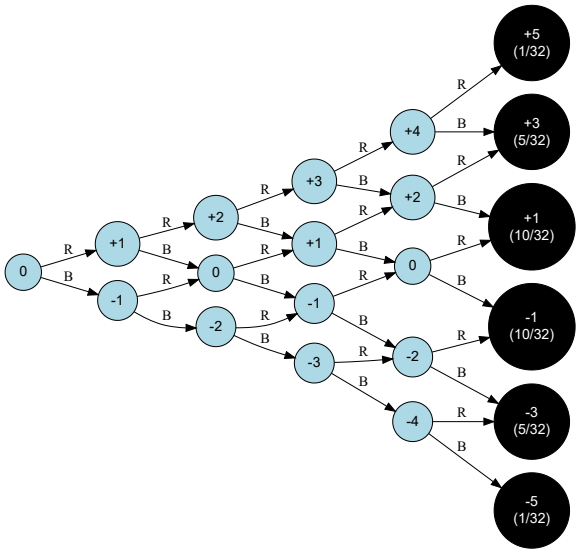
Go to this link: https://aintoha.github.io/DrawCards/ The simulator uses a deck of 52 playing cards (drawn with replacement) to determine the direction of each step. The rules are simple:

- Red Cards (Hearts / Diamonds): Take +1 step forward (Right)
- Black Cards (Spades / Clubs): Take -1 step backward (Left)

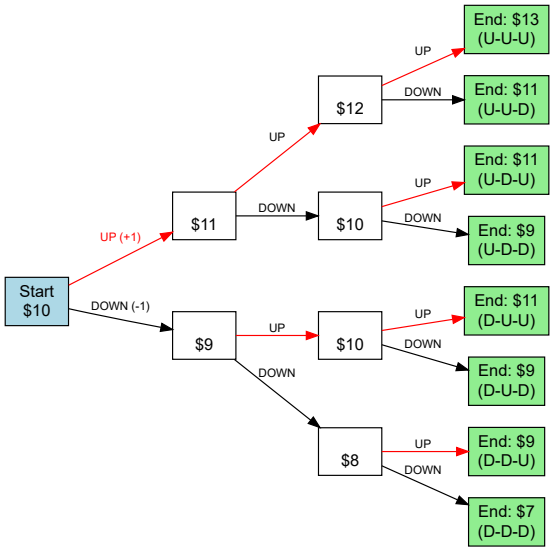
Task: Set the 'Maximum Draws' to 5. Run the simulation 10 times and record your final landing positions. Bring these 10 results to class!

Please answer the following questions on this sheet based on your experience with the simulator.

1. If you set the Maximum Draws to exactly 5, how many unique paths (combinations of Red and Black cards) are possible?
2. In a 5-draw game, what is the exact probability of ending up at position +3? Write your answer as a fraction.
3. In that same 5-draw game, what is the probability of ending at position +3? (Hint: List out the different ways you can draw two Red cards and one Black card).
4. If you take exactly 5 draws, is it possible to land on position +1 or +3? Why or why not?
5. Let the random variable X represent your final position after exactly 5 draws. Construct the PMF table of X.
6. What is the probability that your final position X is strictly greater than 3?
7. What is the probability that your final position X is less than or equal to 4?
8. Suppose you peek at your very first card and see it is Red. Given this condition, what is the probability that your final position after all 5 draws will be exactly +2


0
+1
-1
+2
0
-2
+3
+1
-3
+4
+2
0
-4
+5
(1/32)
+3
(5/32)
+1
(10/32)
-1
(10/32)
-3
(5/32)
-5
(1/32)
R
B

6.0.1 Application of Random Walk



- Drawing a random card from the deck is mathematically identical to clicking the "Take Step" button. Both trigger a single, independent 50/50 random event that dictates the next move.
- The Red Card vs. The Green Upward Line
  - Drawing a Red Card to gain \$10 is the same as a positive market event.
  - On the simulator's Tree Preview, this is represented by the green line moving diagonally upward, increasing the asset's value by whatever number you typed into the "Volatility Step" box.
- The Black Card vs. The Red Downward Line
  - Drawing a Black Card to lose \$10 translates to a negative market shock.
  - On the visual tree, this is the red line moving diagonally downward to a lower price node.
- The 4 Draws vs. The Number of Weeks
  - Playing a game with 4 card draws is identical to setting the simulator's "Number of Weeks" to 4.
  - Each vertical column of circles on the grid represents the exact state of the market after one draw.

Questions

1. If you set the Time Horizon to exactly 4 weeks, how many unique price paths (combinations of Up and Down weeks) are possible?
2. In this 4-week market, what is the exact probability of the stock ending at exactly \$12? Write your answer as a fraction.
3. In this same 4-week market, what is the probability of ending exactly where you started, at \$10? (Hint: List out the different path combinations where the market has exactly two Up weeks and two Down weeks).
4. If you run the simulation for exactly 4 weeks, is it possible for the final stock price to land on exactly \$11 or \$13? Why or why not?
5. Let the random variable X represent the final stock price after exactly 4 weeks. Construct the complete Probability Mass Function (PMF) table for X.
6. What is the probability that your final stock price X is strictly greater than \$10?
7. What is the probability that your final stock price X is less than or equal to \$8?
8. Suppose you watch the market in Week 1 and see the price go UP to \$11. Given this new information, what is the exact probability that your final price after all 4 weeks will end up back at \$10?
9. The 4-week month is over, and you see that the final stock price is exactly \$10. Given this final outcome, what is the exact probability that the market went UP during Week 1?
10. Imagine instead that the month is over and the final stock price is exactly \$12. Given this final outcome, what is the exact probability that the market went DOWN during Week 1? (Hint: Write out the specific paths that lead to a \$12 final price, and look at how they start).